

Fighting in the Shadow of Intervention: A Learned-Proxy Analysis

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Abstract

Anticipated third-party intervention shapes the decision to rebel, yet no measure covers the full population of potential interveners. I construct one: a machine-learning ensemble predicts intervention probability and direction for every dyad-year (1946–2014); a fixed point makes them mutually consistent; their sum is the country-year shadow. Added to a standard onset model, it improves out-of-sample prediction, sharply once resolved by intervener type, and encompasses the leading measures, each limited to one side of intervention or the major powers. In one bargaining framework, three frictions stake predictions on its two axes. Onset falls with the intensity and rises with the tilt toward the opposition—a pattern easier to square with limited commitment or divergent expectations than with the standard screen.

Keywords: civil war, military intervention, machine learning, learned proxy, onset

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A growing body of theory predicts that *anticipated* third-party intervention shapes civil war onset (Cetinyan, 2002; Cunningham, 2016; Dudley, 2024; Gibilisco and Montero, 2022; Kuperman, 2008; Kydd and Straus, 2013; Langø, 2023; Sambanis et al., 2020), yet we lack the measure it calls for: one that covers the full population of potential interveners, varies annually with shifting alliances and foreign-policy alignments, and treats direction—government- versus opposition-biased—as a quantity in its own right. This paper constructs it, and uses it to show that the decision to fight turns on the direction of the anticipated intervention environment and is checked by its intensity.

The construction is the core contribution. A machine-learning ensemble trained on the Regan (2000) dataset of military interventions (1946–2014) predicts, for each directed dyad in a given year, the probability and direction of intervention. Dyad-level predictions are aggregated across all potential interveners to produce a country-year shadow, disciplined by a Nash fixed-point condition requiring each state’s predicted probability to be self-consistent as an input to the others’ predictions.

The measure is built at the level of intervention decisions—who is expected to intervene, and on whose side. Whether it bears on onset is a separate question, and the way to ask it is to carry the shadow into a standard country-year onset logit (Fearon and Laitin, 2003). It earns its place there: added to the structural baseline, the shadow improves out-of-sample prediction—modestly in aggregate, and substantially once resolved by intervener type—and it encompasses the leading existing measures, each capturing only one side of intervention or the major powers. Only once it predicts is there reason to read its direction. The bargaining literature offers three common accounts of why a war both sides could avoid happens anyway: the government’s strength or resolve is private, and the opposition’s demand screens it; the government cannot credibly keep tomorrow the concession it makes today; or the two sides read the odds of intervention itself differently. Worked through a single framework, the three stake different predictions on the shadow’s two axes—the tilt registering which way a war would lean, the intensity the burden it would impose on those who fight it. The estimates lean among these accounts rather than decide between them: onset falls with the intensity in every measurement draw and rises with the tilt in a clear majority, a pattern the standard screening model does not produce and the other two do. The evidence points away from the screen and toward limited commitment or divergent expectations, without separating those two.

The pieces of this account are not new; what has been missing is holding them together. [Cunningham \(2016\)](#) established the government side—the prospect of great-power backing for an incumbent can deter a rebellion outright, shifting the expected balance so far that no challenge pays. [Langø \(2023\)](#), whose model of civil war fought in the shadow of intervention is the most similar to the one developed here, established the opposition side—rebels weigh anticipated outside support, and its prospect emboldens the challenge, even as, pushed far enough, it can induce a government to concede rather than fight. [Gibilisco and Montero \(2022\)](#) brought structural estimation to the question, recovering intervention expectations from an equilibrium among the major powers and finding their effects to run in opposite directions for different powers—a heterogeneity that foreshadows one of the results below. Each holds a piece: one side of intervention, or one set of sponsors—but the ones left out are not a fringe. In these data, fewer than half of the interventions—43 percent—are the work of the major powers; the rest fall to neighbors, coethnics, rivals, and Cold War proxies. None of these accounts holds both sides together across that full population, and the gap is not incidental. Because a country that draws government-biased intervention tends to draw the opposition-biased kind as well, the net tilt of the anticipated environment comes into view only when both sides are measured at once, across everyone, and over time. The field has seen the two forces one at a time; seeing their difference has required measuring them together.

Measuring that environment is the hard part, and it fixes the method. The quantity is an expectation—for every potential intervener, on each side, in each year, the probability that it takes a hand—and no general theory of intervention tells us which states may be dropped from the roster, or which side left out of the ledger. What remains is a large and noisy classification problem, and the task at the measurement stage is to get the probabilities right. This is the natural home of flexible, predictive machinery, and it is the opposite of the choice a structural analysis makes: rather than pare the problem to a set small enough to solve exactly, the learned-proxy approach keeps the whole of it and brings the strongest available predictor to bear, holding interpretation for the second stage ([Knox et al., 2022](#); [Morucci and Spirling, 2024](#)). The structural route is not discarded but embedded—the softmax best response of [Gibilisco and Montero \(2022\)](#) enters the ensemble library as one candidate among many, and the data award it 7.0 percent of the weight, a real but minor share, the flexible learners carrying the rest. Whether that parametric form fits is thus settled empirically rather than

assumed.

The learned-proxy approach has well-documented pitfalls. [Knox et al. \(2022\)](#) catalog these systematically: measurement-stage uncertainty is routinely ignored, proxy quality is asserted rather than tested, and the disconnect between prediction and inference is papered over. This paper follows their recommendations: out-of-fold validation across the 25 imputation draws, measurement uncertainty propagated into Stage 2 by a pairs-cluster bootstrap over all of them, calibration verified by reliability diagrams. The honest accounting is itself a result: 66–87 percent of the coefficient variance originates in the measurement stage, not in primary-analysis sampling—the measurement model, not the regression, is the binding source of uncertainty, precisely the situation [Knox et al. \(2022\)](#) warn about.

The paper proceeds as follows. Section 1 traces three common accounts of bargaining breakdown through a single framework and develops the expectations they stake on the shadow’s two axes. Section 2 constructs and validates the shadow measure. Section 3 turns to onset: it establishes that the shadow predicts out of sample before weighing the mechanisms against one another, then asks which interveners carry the effect.

1 Intervention and the Onset Decision

1.1 The Shadow of Intervention

A rebellion is launched in anticipation of how the world will answer it. Before the first shot, a would-be opposition weighs not only the government in front of it but the outside powers that might join the fight—some on its own behalf, some on the incumbent’s. This is the shadow of intervention ([Cetinyan, 2002](#)), and two of its effects are by now well established. Anticipated *government*-biased intervention deters: a challenger that expects outside backing to make the incumbent all but impossible to defeat sees little gain in fighting, and does not rebel ([Cunningham, 2016](#)). Anticipated *opposition*-biased intervention runs the other way, lowering the expected cost of rebellion and raising its odds of success, and so emboldens the challenge ([Kuperman, 2008](#); [Thyne, 2006](#); [Qiu, 2022](#)).

Why should anticipation touch the decision to fight at all, rather than merely the terms of a settlement? Because a rebellion is not a unilateral leap: a government that sees a credible challenge coming can buy it off with

concessions, so onset is better read as the breakdown of a bargain than as an act taken in isolation (Walter, 2009). Were that bargain always available, the shadow would move the division of the stakes without changing whether war occurs. That it moves onset is the trace of a friction in the bargain, and rather than pick one in advance, this paper traces three common candidates through a single bargaining framework:

- *Private information.* The government’s strength or resolve is its own to know, and the opposition’s demand screens it—the classic risk–return account of war (Fearon, 1995).
- *Limited commitment.* A government cannot credibly promise to honor tomorrow the concession that today’s external threat would wring from it, so a peace that would avert war today is not believed to hold (Walter, 2009; Powell, 2006).
- *Divergent expectations.* The two sides read the shadow itself differently: the government and the rebels need not hold the same estimate of the odds of intervention, and a side that expects help grows optimistic and refuses a settlement it ought to accept (Cunningham, 2016; Cetinyan, 2002).

The three make overlapping but distinct predictions—they converge where any costly-war bargain must, and part exactly where the data can look—and the analysis lets the estimates weigh among them (Section 3).

1.2 Tilt and Intensity

Government- and opposition-biased intervention rarely move one at a time. A country that draws anticipated intervention of one kind tends to draw the other: the features that make outside powers likely to back a government—its strategic value, its alignments, its neighborhood—make them likely to back its challengers too. What distinguishes one anticipated environment from another is therefore less the sheer amount of expected intervention than two things at once: which way it leans, and how heavy it is—its *tilt* toward the opposition and its *intensity*.

The tilt governs the contest. A net lean toward the opposition raises the rebels’ prospects and emboldens them; a net lean toward the government does the reverse—the deterrent and emboldening logics above are the two directions of a single axis.

The intensity governs the burden. Because the two sides move together, a high-intensity environment is one in which a war would draw heavy involvement on *both* sides at once—a competitive, internationalized war. Such wars are the hard ones to end: each added outside stakeholder with its own preferences narrows the range of settlements all parties would accept, and balanced support locks the fighting into stalemate (Cunningham, 2006, 2010; Balch-Lindsay et al., 2008). A challenger who foresees such an environment foresees a long, grinding war, and that prospect deters. Because states in fact intervene where wars are hardest to begin with (Gent, 2008; Gilligan and Sergenti, 2008), it is the *anticipated* environment, not the realized one, that bears on the decision to fight—and anticipated competing intervention is itself associated with longer wars (Akcinaroglu and Radziszewski, 2005).¹

The two forces enter the value of fighting with a built-in asymmetry. On the government side they agree: outside backing for the incumbent both worsens the rebels’ odds and heightens the burden of the war they would fight. On the opposition side they compete: outside backing improves the rebels’ odds but heightens the burden of the very war in which the help would arrive. Whatever the bargain does with these values, then, government-biased expectations should deter, while the opposition channel is a race the war value alone does not resolve—a question the theory hands to the data.

The bargain is what turns war values into onset, and it is where the three mechanisms part company. All three expect onset to fall with the intensity: in a bargain over a costly war, a heavier war leaves more for peace to save, whichever friction is at work. The tilt divides them. Under the screen as written—whatever the government’s private object—the balance of power passes into the demand and out of the decision to fight; no positive tilt effect emerges. Under limited commitment or divergent readings of the shadow, onset rises with the tilt. A positive tilt effect in the data would therefore do more than sit comfortably with bargaining logic; it would point away from the screen—which as written could not have produced it—and toward the other two. Appendix A derives each mechanism’s expectations and tabulates the comparison.

¹These studies estimate the duration of wars that occurred; that a rebel *anticipating* such a war is deterred from starting one is the inference drawn here, not a finding they report.

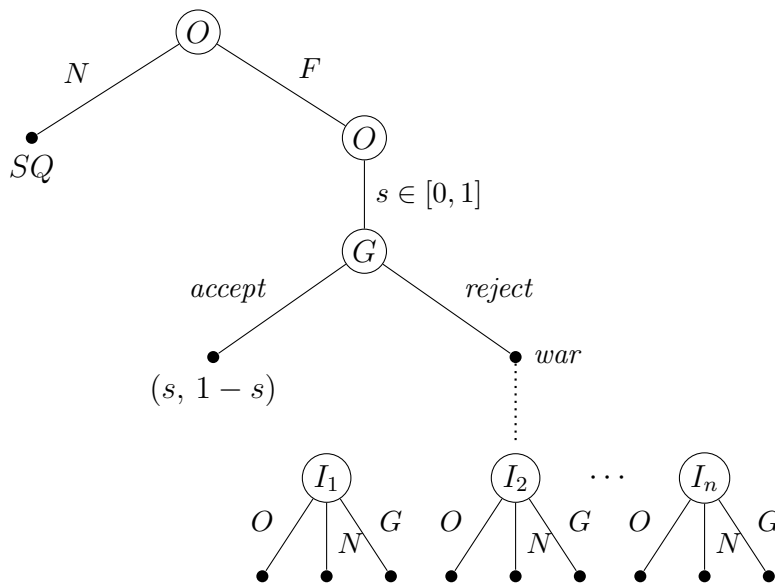


Figure 1: The onset game. The opposition challenges (F) or accepts the status quo (N); a challenge carries a demand s , which the government accepts—a settlement—or rejects, in which case war follows, fought in the environment set by the n potential interveners’ simultaneous choices.

1.3 From Logic to Measurement

The logic is deliberately spare, and one game form carries all three mechanisms. Reduced to its frame, the decision is a would-be rebel weighing whether to challenge in anticipation of how the outside world—every state that might back the government, back the opposition, or stand aside—would answer the war that follows (Figure 1). The mechanisms differ only in what the bargain must survive: private strength or resolve, promises that cannot bind, readings of the shadow that do not agree. The tilt and intensity of the environment are summaries of the anticipated answer, aggregated across everyone who might act; the appendix works the game’s payoffs and equilibria under each friction, and here its structure is enough.

That structure shows what a study of onset must measure. The value of fighting is built from two ingredients—who is expected to intervene, and how much each intervention would matter. Under two simplifying assumptions—that interveners’ contributions enter additively, and that a state expected to stand aside adds nothing—it is a weighted sum over the full roster of

potential interveners,

$$U_O(F; \sigma) = \sum_{i=1}^n [\sigma_i(O) u_O^i(O) + \sigma_i(G) u_O^i(G)] , \quad (1)$$

one term for each of the two unknowns. The intervention probabilities σ_i are a prediction problem—for every potential intervener, the probability and direction of its involvement—and the work of Section 2, where flexible, data-driven methods earn their place. The utility weights u_O^i are recoverable only from the onset behavior the expectations are meant to explain, and are estimated in Section 3. The measure is built in the first stage; its bearing on onset—and with it, the standing of the mechanisms—is read in the second.

2 A Learned Proxy for the Shadow of Intervention

Operationalizing the shadow requires a measure of the expected intervention environment at the moment a potential rebel group weighs the costs and benefits of open conflict. The construction proceeds in two stages: a directed-dyad classifier estimates intervention probabilities for every potential intervener, and a country-year aggregation sums them under a Nash fixed-point condition ensuring self-consistency.

2.1 Outcome Variable

The Stage 1 training labels draw primarily from [Regan \(2000\)](#), whose dataset records third-party interventions in civil conflicts from 1944 to 1999. I extend the coding through 2014 by hand-coding unambiguous post-1999 military interventions—personnel-based only: troops, combat advisors, or directed proxy forces; arms, aid, sanctuary, and UN peacekeeping are insufficient—adding 32 directed dyad-year records across 14 hosts, for a combined table of 456 records (1944–2014; coding and sources in [Online Appendix B](#)).

The civil war universe is drawn from the Correlates of War Intra-State War Dataset (v5.1) ([Dixon and Sarkees, 2016](#)), restricting to wars fought for central control (type 4) or local/secessionist issues (type 5). The COW threshold ($\geq 1,000$ battle deaths) is theoretically appropriate: the mechanism requires rebels organized enough to form rational expectations and con-

flicts large enough that outside military involvement is a realistic prospect.² This yields 153 distinct onset country-years between 1946 and 1999 and 191 through 2014. I restrict attention to *military* interventions, coded as *government-biased* (1) or *opposition-biased* (2).

For each directed dyad ($B \rightarrow A$) active in a civil war onset year, the training label is: 0 (no military intervention), 1 (government-biased), or 2 (opposition-biased). Non-onset dyad-years are excluded from the training set. This produces 211 intervention-coded onset observations—125 government-biased and 86 opposition-biased—across 50 host countries and 57 unique intervening states. The five permanent Security Council members account for 90 of these events (43 percent); the remaining states—Cold War proxies, neighbors, regional patrons, and coethnics—account for the majority (Table 1).

Training exclusively on onset country-years is not a convenience—it is the only option. Intervention in a civil war is undefined absent a civil war; in non-onset years the outcome is not “zero intervention” but rather unobserved. The consequence is that the training sample is selected on precisely the outcome that the Stage 2 analysis tries to explain: if the shadow deters onset, then high-shadow country-years are systematically underrepresented among onsets, and the classifier learns the feature–intervention mapping from an endogenously filtered sample. There is no design that avoids this. The defense rests on what the classifier is asked to learn. The training target is the conditional relationship between *dyad-level* features—alliance portfolios, capability ratios, colonial ties, geographic proximity, regime similarity—and the direction of military intervention, given that a war has started. These bilateral characteristics are plausible determinants of how state B behaves toward state A in a conflict regardless of what brought that conflict about. The fixed-point iteration then handles the extrapolation: it asks what the equilibrium intervention environment *would* look like in any country-year, including those where onset was never observed, without requiring that the training sample be representative of the full population.

²The UCDP/PRIO Armed Conflict Dataset offers broader, more recent coverage and a more detailed external-support record; extending the learned-proxy approach to it is a natural next step (sketched in the conclusion), though its lower (≥ 25 -death) threshold—admitting many smaller conflicts in which intervention is rarer—would leave a sparser training signal.

State	Gov.	Opp.	Total
United States [†]	29	12	41
USSR / Russia [†]	14	5	19
France [†]	9	4	13
United Kingdom [†]	7	4	11
Libya	1	6	7
China [†]	2	4	6
Cuba	5	1	6
Iran	2	4	6
Uganda	3	2	5
Vietnam	1	4	5

Table 1: Most frequent military interveners, 1946–2014 (states with fewer than five interventions omitted). [†] UN Security Council P5 member. Gov. and Opp. from the Regan target coding and the post-1999 extension.

2.2 Feature Set

The classifier draws on 109 directed-dyad-year features in four families: characteristics of both the potential intervener and the host—capabilities, regime, power status, and host-side conflict history and social structure; bilateral ties—alliances, trade, proximity, colonial history, disputes and their expected outcomes; foreign-policy alignment—ideal-point distances and shared IGO memberships; and 12 spatial lags capturing the weighted behavior of the other potential interveners in the same host-year (Online Appendix E). Every time-varying feature enters at its value in the year preceding the onset, on both sides of the dyad: the classifier’s information set closes before the onset year begins. Many variables enter twice—once for the intervener, once for the host—so the classifier can learn asymmetric effects; every variable, with sources and definitions, appears in Online Appendix D. Missing data are multiply imputed (five country-year $\times$ five undirected-dyad imputations, yielding 25 complete datasets; Online Appendix C), and the features are standardized and reduced by principal components to 90% of cumulative variance (56–57 components across draws).

2.3 Classifier Design

The multinomial outcome (0 = no intervention, 1 = government-biased, 2 = opposition-biased) is modeled using a super-learner ensemble (van der Laan et al., 2007).

Nine component classifiers spanning three model families are trained within a ten-fold cross-validation scheme: two tree ensembles (a random forest and two histogram-based gradient boosting specifications at different learning rates), four logistic regression variants (ridge, elastic-net, lasso, and unpenalized multinomial), and two multi-layer perceptrons (shallow and deeper architectures). The unpenalized multinomial logit is the softmax best response of structural intervention games (Gibilisco and Montero, 2022), so the parametric benchmark competes inside the library on out-of-fold performance. Each classifier produces a vector of class probabilities for every training observation. The super learner combines these using non-negative least squares (NNLS) weights chosen to minimize held-out multinomial log-loss, so that the ensemble adaptively up-weights component classifiers that predict well on out-of-fold observations.

All reported performance statistics are based on out-of-fold predictions to guard against overfitting—following Wang’s (2019) demonstration that evaluation of class-imbalanced political-event models on training data produces severely inflated performance estimates (Muchlinski et al., 2016). Because intervention is rare, the metric matters: the primary summary is the proportional reduction in log-loss against the class-frequency null (PRL), a proper scoring rule; ROC-AUC is reported for comparability but is insensitive to class imbalance, so the area under the precision–recall curve (AUC-PR)—whose no-skill baseline is the prevalence itself—accompanies it.

Spatial lags and Nash fixed-point iteration. The broader motivation for the two-stage design is that intervention decisions are endogenous to the conflict environment: interveners select into conflicts strategically, and a single-equation model that treats realized intervention as a covariate in an onset regression conflates the causal effect with the selection process that generated it (Signorino, 2002; Gent, 2008); rebel groups in Bosnia and Kosovo conditioned onset on anticipated support, so realized intervention is partly a consequence of expectations rather than an independent cause (Kuperman, 2008). Predicted expectations, not realized intervention, are the appropriate regressor.

The spatial context variables of Appendix E depend on the intervention choices of *other* potential interveners—quantities that are themselves to be predicted, and consequentially so: when one state backs a government, rival states face a selective incentive to back the opposition (Salehyan et al., 2011; Findley and Teo, 2006), and it is the alignment among interveners, not merely their count, that carries the signal (Aydin and Regan, 2012). Without an explicit consistency requirement, spatial lags computed from *observed* training-data choices embed a deterministic discrepancy between lag inputs and probability outputs whenever the model is applied out of sample—systematic error, not sampling noise, in a measure built to capture counterfactual environments.

The fix is to require that the predictions be self-consistent inputs to themselves:

$$\sigma^*(B, A, t) = \hat{M}(X(B, A, t, \sigma^*)) \quad \forall (B, A, t), \quad (2)$$

where $\hat{M}$ is the trained super-learner classifier. At σ^* , the spatial lags are derived from the model’s own equilibrium predictions, not from historical observations. The deterministic discrepancy disappears by construction.

This self-consistency condition is also the Nash equilibrium condition for a natural class of games. The primitive is the best-response correspondence, not the game: a game specifies one *derivatively*, via utility maximization, while $\hat{M}$ estimates one *directly*, from the data—and Nash equilibrium is a fixed point of that correspondence, whichever payoffs generate it. σ^* is thus the equilibrium of every game whose best-response correspondence $\hat{M}$ estimates.³ If the training data reflect equilibrium play and the lags are

³The set of rationalizing payoff functions for any interior σ^* is non-empty—choose utilities that make each player indifferent among actions assigned positive probability. What is identified is the correspondence; the payoffs that generate it are not. A natural parametric fiber of these classes is the linear-utility family $u_i(a, \sigma_{-i}) = \alpha \cdot f_i(a) + \beta \cdot W_i(a, \sigma_{-i})$, where f_i captures state-specific intervention incentives and W_i the spatial interaction term. The best-response from this family is a softmax—the functional form of the multinomial logit component of the super-learner—so the parametric case is explicitly represented in the ensemble while the broader non-parametric mixture accommodates departures from it. The iterative procedure that solves Equation (2) is the nonparametric analogue of the nested pseudo-likelihood (NPL) algorithm of Aguirregabiria and Mira (2007): the fitted super-learner replaces the parametric best-response function, and the fixed-point iteration updates beliefs (spatial lags) until self-consistency, as in A&M’s two-step PML. In the parametric case, Crisman-Cox and Gibilisco (2021) show that this class of iterative estimators substantially outperforms direct maximum likelihood estimation

Model	PRL	ROC-AUC	AUC-PR	Log-loss
Super-learner	39.6% (0.8%)	0.958 (0.004)	0.32 (0.01)	0.032 (<0.001)
Random forest	8.3% (6.7%)	0.936 (0.006)	0.27 (0.02)	0.048 (0.004)
Lasso logit	31.0% (0.8%)	0.936 (0.003)	0.22 (0.01)	0.036 (<0.001)

Table 2: Stage 1 classifier performance (out-of-fold; mean and SD across the 25 draws) for the super-learner, its top-weighted nonlinear component, and the best logit. AUC-PR is for any intervention (no-skill baseline ≈ 0.008); ROC-AUC is macro-averaged. Full components in Table 13.

sufficient statistics for the strategically relevant information in σ_{-i} , then σ^* estimates the fixed point of the true best-response correspondence: Equation (2) corrects a concrete measurement error and identifies the equilibrium of the underlying intervention game at the same time.

Equation (2) has no closed form; it is solved by fixed-point iteration in two stages: five warm-start passes during training, then, with the classifier held fixed, prediction-stage updates of the spatial lags to convergence—for every directed dyad-year, onset and non-onset alike, so the shadow reflects the counterfactual equilibrium in any country-year (Appendix F.3).

2.4 Predictive Performance

Table 2 reports out-of-fold performance for the super-learner and two selected components: the lasso logistic regression (the best single component by standalone PRL) and the random forest (which despite lower standalone PRL receives the highest NNLS weight at 53.0%, reflecting complementary predictive information). Full results for all nine component learners appear in Appendix F, Table 13. All statistics average across the 25 complete datasets, with standard deviations across imputation draws in parentheses.

The ensemble achieves a PRL of 39.6% and a macro-average ROC-AUC of 0.958. For any intervention, the area under the precision–recall curve reaches 0.32: averaged over the recall range, roughly a third of the dyads the classifier ranks as likely interveners are genuine interventions, versus fewer than one

of strategic games under equilibrium multiplicity. Uniqueness of the fixed point is not guaranteed in general; empirically the prediction-stage iteration converges for 23 of the 25 imputation draws within tolerance (mean 13 passes), the remainder plateauing just above it (Appendix F.3).

in a hundred (0.008) at the no-skill baseline. A PRL of 39.6% means the classifier recovers nearly two-fifths of the information the class-frequency null discards—substantial for a three-class problem whose modal outcome exceeds 99% of observations. The natural worry is that it earns this by calibrating the hay—sharpening the abundant non-intervention predictions—rather than finding the needles the shadow requires. Decomposing the log-loss reduction by true class rules that out: 98 percent of the PRL originates in the 0.8 percent of observations where intervention actually occurred, only 2 percent in improved calibration of non-events.

Because the classifier sees only publicly observable characteristics, 39.6% is a lower bound on how foreseeable the intervention environment is to the rebels themselves, who hold private intelligence and patron communications besides—the empirical footing of the rational-expectations reading (Appendix A, *Remark: rational expectations*).

The gap between the ensemble and the best single logistic specification—39.6% versus 31.0% PRL on an identical feature set—reveals nonlinear structure that curated political-science feature sets typically lack (Morucci and Spirling, 2024); a measurement-grade feature space is the exception that rewards flexible learners. The NNLS weights say so directly: ensemble mass concentrates on the nonlinear learners—the random forest (53.0%), the perceptrons (20.2%), and gradient boosting (17.3%)—while the multinomial logit takes 7.0% and the penalized logistic variants, despite the highest standalone PRL, receive near-zero weight (Appendix F, Table 13). With the logistic family competing inside the library and down-weighted by the data, the ensemble’s advantage is not an artifact of architecture choice.

Because Stage 1 probabilities are subsequently summed across approximately 135 potential interveners, systematic miscalibration compounds; reliability diagrams confirming calibration for each outcome class appear in Appendix F. These diagnostics follow Knox et al.’s (2022) recommendation to “always assess and report measurement performance” (their §5.4): the PRL and AUC quantify predictive accuracy, and the reliability diagrams verify the calibration assumptions on which the aggregation to country-year expectations depends.

2.5 Aggregating to Country-Year Intervention Expectations

The Stage 1 classifier produces, for each directed dyad ($B \rightarrow A$) in a civil war onset country-year, probability estimates $\hat{\sigma}_B(1)$ and $\hat{\sigma}_B(2)$ for government- and opposition-biased intervention, respectively. The country-year intervention expectation for host A in year t sums these across the full roster of potential interveners, one total for each side:

$$E_A^G(t) = \sum_B \hat{\sigma}_B(1), \quad E_A^O(t) = \sum_B \hat{\sigma}_B(2).$$

Both sums are mechanically larger in later years as the system expands. I include a year trend in all second-stage regressions to absorb this secular growth. The sums are also right-skewed; following [Burbidge et al. \(1988\)](#), I apply the inverse hyperbolic sine (asinh) transformation before entering them into the onset models.

2.6 Properties of the Shadow Measure

The measure behaves as the theory requires; Online Appendix [G](#) documents it: the highest-probability dyads recover the historically correct patrons on the correct sides in Angola, Ethiopia, and Afghanistan; the top-ranked opposition-biased predictions are dominated by non-P5 interveners that a five-power framework would miss; the distributions are heavily right-skewed; and E^G correlates moderately with the Lake hierarchy ($r = 0.29$) and the [Gibilisco and Montero \(2022\)](#) structural estimates ($r = 0.47$).

3 The Decision to Fight

Measure in hand, I study whether intervention expectations affect the decision to initiate civil war. The *Entrants* model augments a standard onset logit with the two shadow variables constructed in the previous section:

$$U_O(F; \hat{\sigma}) = \mathbf{X}\beta + \gamma^O \sum_i \hat{\sigma}_i(O) + \gamma^G \sum_i \hat{\sigma}_i(G) + \epsilon,$$

where $\mathbf{X}$ contains the baseline country-year predictors and the two summations are the expected total number of opposition-biased and government-

biased interveners, respectively.⁴ This is the simplest specification of Equation (1), in which all potential interveners carry the same utility weight ($u_O^i(O) = \gamma^O$ and $u_O^i(G) = \gamma^G$ for all i); the heterogeneous-utilities extensions below relax it.

The two-stage design extends the logic of statistical backwards induction (Bas et al., 2008) to an n -player environment: the intervention subgame involves up to 160 simultaneous movers, separability and the Nash fixed-point reduce it to one in which each intervener’s marginal probabilities suffice, and Stage 2 recovers the opposition’s utility parameters from their aggregate effect on onset.

3.1 Empirical Strategy

Whether intervention expectations shape onset is, in the first instance, a question of model comparison. Following Clarke (2007), the relevant test is not whether the shadow variables are individually distinguishable from zero—for a generated and internally collinear regressor, the conventional apparatus is ill-suited to that question⁵—but whether a model that includes them predicts better, out of sample, than one that does not. The worth of a variable is then measured by what its removal costs: I compare the out-of-sample fit of the onset model with and without the shadow—leave-one-covariate-out importance (Lei et al., 2018), the reduced-model logic that McAlexander and Mentch (2020) bring to onset prediction—applied here across the whole covariate set. A measure that improves on a purely structural forecast—even modestly, since *out-of-sample* fit is the one signature overfitting cannot counterfeit—earns a reading of the direction in which it enters. This is a weaker standard than significance, and a more honest one for a learned proxy: it asks the shadow to carry information a held-out fore-

⁴In estimation the summations enter through the inverse hyperbolic sine transform for variance stabilization—a concavity the theory neither implies nor rules out (its content is the pair of axis signs, Section 1). Because asinh is strictly increasing, coefficient signs are invariant to the choice, and the out-of-sample comparisons do not turn on it; the exact shape of the utility function is not identified from these data.

⁵The two channels are nearly the same variable in these data: E^G and E^O correlate at $r \approx 0.94$, because a country that draws one kind of anticipated intervention tends to draw the other. An individual coefficient asks how onset moves with one channel while the other is held fixed—a comparison the data almost never supply—so individual standard errors are large and unstable whatever the estimator. Section 3.3 works instead in the two combinations the data do identify: the sum of the channels and their difference.

cast can use, not to clear a threshold its generated and collinear standard errors were never built to certify.

The rival is the *Baseline*: a logistic onset model on the structural covariates of [Fearon and Laitin \(2003\)](#)—prior war, lagged log income and population, mountainous terrain, oil, new-state status, political instability, ethnic and religious fractionalization, and democracy—the specification the onset literature has long treated as its reference. This is not a replication of Fearon and Laitin but their covariates estimated on the present sample (COW Intra-State Wars v5.1, 1946–2014) with current sources: income and population from V-Dem v15 ([V-Dem Institute, 2025](#)), democracy from Polity, conflict history from COW, and the fixed structural characteristics from the Fearon–Laitin replication data. Time-varying covariates are lagged one year against post-onset contamination, and a year trend absorbs the mechanical growth of the summation-based shadow variables as the state system expands. The *Entrants* model adds the two shadow variables to this Baseline; everything that follows asks what they add.

Three questions organize what follows. First, how much does the shadow improve out-of-sample prediction, and how does its contribution compare with the structural predictors alongside it? I assess this by leave-one-country-out cross-validation—all of a country’s observations held out together. Because onset is rare, I report the proportional reduction in log-loss and the area under the precision–recall curve, each against its no-information baseline, under both a logistic specification and a random forest, so the comparison does not turn on functional form. Second, in what direction does the measure enter? The mechanisms of [Section 1](#) stake different predictions on the two axes the data can identify, and I read the sign structure in the register [Gill \(1999\)](#) and [Ward et al. \(2010\)](#) urge for measurement-based claims—direction and out-of-sample behavior in place of the asterisk. Third, which interveners carry the effect—does resolving the shadow by intervener type strengthen the prediction and recover structure the single summed expectation conceals?

Two features of the design demand that uncertainty be handled with care. The shadow is estimated at Stage 1, not observed, so treating it as data would understate Stage 2 uncertainty; and its two channels are collinear, more than the individual coefficients can be asked to separate. I retain all 25 multiply-imputed shadow measures and propagate their spread by the $T \times P$ procedure of [Knox et al. \(2022\)](#): the analysis runs on each of the $T = 25$ measurement draws, bootstraps over resampled countries within each, and pools the resulting coefficient vectors. Point estimates of fit average the held-

out predictions across draws; every interval reflects the full measurement spread. This is the discipline a learned proxy demands: its reach must be carried into inference honestly, uncertainty and all.

3.2 Prediction and Subsumption

Begin with the shadow’s predictive worth. Across the onset model’s full covariate set, no variable’s absence degrades out-of-sample fit more than the shadow’s (Table 3): removing both shadow variables lowers the out-of-sample proportional reduction in log-loss further than removing any structural predictor, prior war included, and the government channel alone outranks every canonical covariate. In a rare-event onset model most structural predictors buy little out of sample; against that backdrop the shadow buys the most. The [Mentch and Hooker \(2016\)](#) prediction-difference test agrees jointly at every configuration, its permutation counterpart in every measurement draw (Appendix I).

The levels agree. Onset is rare—184 onsets in 8,835 country-years, a no-skill precision–recall baseline of 0.021—and adding the two aggregate shadow variables to the structural *Baseline* raises out-of-sample AUC-PR from 0.057 to 0.074 under leave-one-country-out cross-validation, a paired improvement whose bootstrap interval excludes parity (Table 4). The aggregate gain is modest; it grows to more than double the baseline once the shadow is resolved by intervener type (Section 3.4). It holds under both a logit and a random forest.

Neither of the field’s existing proxies for anticipated intervention was built to predict onset: Cunningham’s serves a hypothesis test about hierarchy and deterrence, and the payoff of the Gibilisco–Montero structural estimates is counterfactual analysis, which a predictive measure does not supply. The question Table 5 asks is narrower—for the onset problem, does the shadow hold the information they hold? On each proxy’s coverage subsample, adding the Cunningham security-hierarchy measure or the Gibilisco–Montero P5 estimates to the shadow leaves out-of-sample fit unchanged, while adding the shadow to either improves it substantially—a gap that widens sharply once the shadow is resolved by type (Section 3.4).

Dropped from <i>Entrants</i>	Δ PRL	95% interval	Δ AUC-PR	RF Δ AUC-PR
Both shadows (E^G , E^O)	+0.032	[+0.017, +0.047]	+0.017	+0.016
E^G (gov. shadow)	+0.018	[+0.007, +0.030]	+0.012	+0.008
E^O (opp. shadow)	+0.002	[+0.001, +0.004]	+0.002	+0.005
Prior war	+0.011	[−0.001, +0.023]	+0.009	−0.005
Instability	+0.007	[−0.001, +0.017]	+0.003	−0.004
Log income	+0.005	[−0.006, +0.015]	+0.001	−0.002
New state	+0.004	[−0.004, +0.013]	+0.005	+0.002
Democracy	+0.002	[−0.004, +0.008]	+0.002	+0.000
Rel. frac.	+0.001	[−0.007, +0.009]	−0.002	−0.005
Noncontiguous	+0.001	[−0.007, +0.008]	−0.000	−0.003
Log pop	+0.001	[−0.004, +0.005]	−0.001	+0.002
Eth. frac.	+0.000	[−0.005, +0.005]	+0.002	+0.001
Mountains	+0.000	[−0.005, +0.005]	−0.002	−0.002
Oil	−0.001	[−0.005, +0.002]	−0.000	−0.001
Year trend	−0.001	[−0.003, +0.001]	−0.000	−0.000

Table 3: Out-of-sample drop-column (leave-one-covariate-out) importance for the *Entrants* onset model: the degradation in out-of-sample fit when each covariate is removed and the model refit (Lei et al., 2018), under leave-one-country-out cross-validation. Larger Δ = more important. Intervals are 95% paired country-cluster bootstrap (differences taken within each resample, $B = 1,000$). Final column: the same exercise refit with a random forest.

3.3 Direction

I turn to the direction in which the shadow enters. The three mechanisms of Section 1 enter with overlapping bets: all three expect onset to fall with the common intensity of the anticipated environment; commitment and divergent expectations expect it to rise with the net tilt toward the opposition, where the common-prior screen does not; and all agree that government-biased expectations deter, while the raw opposition channel is left unsigned—a race between the better odds outside backing brings and the heavier war it portends. The raw estimates behave accordingly: government-biased expectations enter negative in 96 percent of the 25 measurement draws, the opposition channel positive in 76 percent and fragile (Table 6). Their individual magnitudes cannot be trusted—with the two channels nearly collinear ($r \approx 0.94$), the individual coefficients swing from one measurement draw to the next.

What the data can speak to is not either channel alone but the two directions in which the pair moves: *together*—deeper into the shadow of both kinds of intervention at once—and *apart*—the balance tilting toward one side. These are the sum and the difference of the channels, a *common intensity* $E^O + E^G$ and a *net tilt* $E^O - E^G$, and they are precisely the coordinates in which Section 1 stated its predictions: the intensity that moves the burden, the tilt that moves the contest. Both are well identified where the individual channels are not, and the two combinations the data can estimate are exactly the two on which the mechanisms disagree.

Read in these terms, the intensity effect is unambiguous and the tilt effect is real but softer. The common intensity enters negative in *every* one of the 25 measurement draws, its pooled interval lying entirely below zero: a heavier anticipated environment, of either kind, lowers the risk of onset. The net tilt enters positive in 84 percent of draws—a clear majority, though its pooled interval still straddles zero (Table 6). I lean on the signs rather than the magnitudes by necessity: between 66 and 87 percent of each coefficient’s variance is the shadow’s own measurement error, carried from Stage 1 and invisible to the conventional standard error. Both moves are large: for a country at the sample mean, shifting the intensity from its tenth to its ninetieth percentile lowers predicted onset from 1.94 to 0.22 percent (Table 6).

The pattern is suggestive about mechanism, though it does not decide. The intensity effect is common property of any costly-war bargain—it corroborates the burden the theory attaches to intervention without selecting among the frictions. The tilt is the discriminating axis: a positive tilt cannot arise through a frictionless bargain, and the common-prior screen as written does not produce it either, whatever the government’s private object (Appendix A), so onset rising with the tilt sits more comfortably with limited commitment or divergent readings of the shadow than with the screen. The evidence for the tilt is a clear-majority sign rather than a sharp interval, so I read it as leaning against the screen, not ruling it out. The firmer trace of the opposition mechanism comes not from the aggregate tilt but from resolving the shadow by intervener type, to which I turn next.

3.4 Heterogeneous Utilities

The Entrants model weights every potential intervener equally. The theory does not require that: letting the opposition’s valuation of an intervener

vary with the intervener’s characteristics produces, on aggregation, a type-weighted shadow term for each attribute. I resolve the shadow along six intervener attributes—major-power status (*Powers*), contiguity to the host (*Neighbors*), a shared dominant ethnic group (*Coethnics*), a former colonial tie (*Rulers*), bilateral rivalry (*Rivals*), and the local military balance—and ask what the type structure adds.

It adds a great deal, and the aggregate’s modest predictive gain becomes a large one. Resolving the shadow by type raises out-of-sample AUC-PR to 0.143—2.5 times the structural baseline, and nearly double the aggregate shadow (Table 4). The gain rewards structure, not flexibility: the type-weighted logit out-predicts a random forest on the same inputs (0.080), so the theory’s separate-weight-per-attribute form carries the content, not black-box curvature. The disaggregation lifts both channels (Table 7): resolving the government channel by type raises its AUC-PR from 0.07 to 0.10, the opposition channel from 0.06 to 0.10. The opposition shadow, the weaker arm of Section 3.3 in aggregate, gains the most, closing to near-parity with the government side once the interveners behind it are held apart.

Resolved this way, the shadow also dominates the existing proxies outright. On the Cunningham and Gibilisco–Montero coverage subsamples the full type-resolved model reaches AUC-PR 0.151 and 0.156, more than double the baseline augmented with each proxy, which adds nothing to the full shadow (Table 5). The proxies measure a corner of what the shadow measures.

Which interveners, and in which direction? Prediction gates the reading: I take the direction of a type only where resolving the shadow by that type improves prediction over the aggregate. Three types clear that bar—neighbors (0.09), major powers (0.09), and rivals (0.09)—and for those three the per-type net tilt is stable across the measurement draws (Table 8). Down the opposition column the effect is not one thing: expected opposition-biased intervention by a neighbor or a rival *emboldens*, while by a major power it *deters* (Table 8). The government side, where it predicts, deters for every type—the one per-type sign the theory fixes, no intervener emboldening from the government’s side, and no type contradicts it.

The opposition is emboldened, then, by the interveners with a proximate, adversarial stake in the conflict—the neighbor and the rival, whose backing a would-be rebel can read as committed⁶—while the prospect of great-power

⁶The rival channel has a natural micro-foundation in Qiu (2022), who shows that states

sponsorship cuts the other way—in the model’s terms, the signature of an intervener whose anticipated involvement would internationalize the war more than tilt it ($k'b_i > \pi'w_i$; Appendix A). Why great-power involvement carries that weight I leave open. But the reversal resolves the puzzle Section 3.3 left open: the aggregate opposition shadow looked weak because it was summing effects of opposite sign. The emboldening was never absent from the data; it was averaged away, resident in the composition of who is expected to intervene rather than in a single summed expectation.

4 Conclusion

Civil war does not begin in a vacuum. Opposition groups weigh the prospect of outside military assistance against the threat of international resistance before they decide to fight. I have argued that these anticipatory calculations—the shadow of intervention—systematically shape the decision to fight, and I have developed a theory, a measure, and an evidentiary case for that claim.

The theoretical framework embeds the onset decision in a bargain fought in the shadow of up to n potential outside powers. Three assumptions—separability across interveners, non-intervention as the reference point, and a contest-burden decomposition of the war value—reduce the opposition’s calculus to a weighted sum of intervener-specific probabilities, moving on two axes: the net tilt of the anticipated environment and its intensity. Worked under the common bargaining frictions, the model signs the axes and the government channel, declines to sign the raw opposition channel, and fixes what the evidence can discriminate: intensity deters in every costly-war bargain, while an emboldening tilt requires limited commitment or divergent expectations. The estimates then lean against the screening account as written and toward commitment and divergent expectations, without deciding between the two. The framework generalizes prior work by treating both directions of intervention, the full population of potential interveners, and their strategic interdependence.

Operationalizing this framework required constructing a measure that did not yet exist. The Stage 1 classifier turns a rich directed-dyad feature

sponsor insurgents in their adversaries to drain them for leverage in interstate disputes—giving a rival a standing interest in a rebellion’s survival, and a would-be challenger a backer it can count on. What that model argues from cases, the shadow recovers as an aggregate regularity across the full population of interveners.

set into calibrated intervention probabilities for every potential intervener, and multiple imputation across 25 complete datasets carries measurement uncertainty through to the final estimates.

The onset analysis returns three findings. First, the shadow predicts. Added to a structural onset model, the two aggregate shadow variables improve out-of-sample precision under leave-one-country-out cross-validation—modestly on their own (the area under the precision–recall curve rises from 0.057 to 0.074) and to more than double the baseline once the shadow is resolved by intervener type (to 0.143)—and by out-of-sample drop-column importance the shadow is the model’s single most valuable component, above every structural covariate. It also subsumes the existing proxies for anticipated intervention: neither the Lake security hierarchy (Cunningham, 2016) nor the Gibilisco and Montero (2022) structural estimates adds predictive content once the shadow is included. Second, the shadow enters onset along the two directions on which the mechanisms disagree: onset falls with the overall intensity of expectations in every measurement draw and rises with the net tilt toward the opposition in a clear majority—the two axes the collinear channels ($r \approx 0.94$) leave identified. The pattern leans away from the standard common-prior screen and toward limited commitment or divergent expectations, without deciding between them; the government-biased channel, on which contest and burden agree, is the unambiguously deterrent one. Third, the opposition effect, weak as a single summed expectation, is concentrated in particular interveners and reverses sign across them: opposition backing by a neighbor or a rival emboldens, while backing by a major power deters—a reversal I do not press to explain here. The undifferentiated measure was averaging these offsetting effects away.

The results carry implications beyond the civil war literature. The two-stage strategy introduced here—predict the probabilities of downstream actions, then use those predictions as regressors in a model of the initiating decision—applies wherever a strategic choice is made in anticipation of others’ responses. Within the civil war literature, it supplies what the macro-correlates framework cannot: intervention expectations shift with the decisions of specific leaders and governments—year to year, conflict to conflict—and so carry the temporal variation that slow-moving structural risk factors lack.

The design also speaks to the debate over machine learning’s place in quantitative political science. Grimmer et al. (2021) distinguish measurement from causal inference and name their conflation the cardinal sin; the

two stages here keep them apart—an agnostic ensemble evaluated on predictive criteria, then a parametric onset model carrying the inferential weight. [Knox et al. \(2022\)](#) close the loop: once a proxy is learned, its uncertainty must be propagated and its imperfections shown not to distort conclusions. Under their “average monotonicity” condition (§3.1)—more of the true concept implies more of the proxy, which the Stage 1 calibration verifies—the estimated signs of the tilt and intensity effects are valid even if the magnitudes are attenuated.

The two stages also return opposite verdicts on flexibility: the agnostic ensemble wins Stage 1, where tens of thousands of dyad-years and near-deterministic exclusion zones present exactly the jagged, interaction-heavy mapping tree-based learners are built for ([Montgomery and Olivella, 2018](#)), while the theory-shaped logit wins Stage 2, where onsets are rare, the index is the theory’s, and flexibility buys variance with nothing left to find. The division is the lesson: flexibility earns its keep where data are plentiful and the truth is jagged; structure, where events are rare and the theory supplies the functional form.

One limitation is worth stating plainly. The analysis rests on the Correlates of War civil war universe, the [Regan \(2000\)](#) intervention data, and an onset specification in the [Fearon and Laitin \(2003\)](#) tradition—sources that were standard when much of this literature was built but are now dated relative to the UCDP/PRIO Armed Conflict Dataset ([Gleditsch et al., 2002](#)) that anchors contemporary civil war research. The ACD covers a broader conflict universe and a more recent period, and its External Support Dataset ([Meier et al., 2023](#)) codes third-party involvement at a finer, actor-level granularity than the interstate intervention record used here. The measurement strategy this paper develops is not tied to these particular sources; the estimates it produces are.

Extending the approach to that data universe is a natural next step, and the two-stage design ports directly. Stage 1 would train the same super-learner on ACD onset dyads, drawing intervention labels from the External Support Dataset—whose disaggregation by support type (troops, materiel, sanctuary, funding) would permit a richer shadow: the anticipated *forms* of involvement, not only its bias. The Nash fixed point and the country-year aggregation carry over unchanged; Stage 2 would regress a contemporary onset specification on the resulting expectations, in place of the [Fearon and Laitin \(2003\)](#)-style model used here. Two features of the ACD would sharpen the exercise: its lower battle-death threshold admits many smaller conflicts

in which intervention is rarer, placing a premium on rare-event metrics; and its currency through the 2010s and 2020s would bring the Syrian, Libyan, Sahelian, and Ukrainian intervention environments within reach. I leave this to future work.

Whether anticipated intervention shapes the decision to fight is a substantive question; that it can be measured—at scale, across time, and across the full population of potential outside powers—is the prior claim on which the answer depends.

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	PRL	ROC-AUC	AUC-PR
<i>Logit</i>			
Baseline	0.079 [0.042, 0.118]	0.746 [0.693, 0.800]	0.057 [0.044, 0.076]
Entrants	0.111 [0.076, 0.147]	0.776 [0.733, 0.821]	0.074 [0.053, 0.116]
Full	0.192 [0.155, 0.232]	0.852 [0.819, 0.884]	0.143 [0.104, 0.212]
AUC-PR lift, paired			1.3× [1.1, 1.7]
AUC-PR lift (Full), paired			2.5× [2.0, 3.4]
<i>Random forest</i>			
Baseline	−0.095 [−0.193, −0.018]	0.712 [0.659, 0.762]	0.053 [0.036, 0.076]
Entrants	0.007 [−0.072, 0.060]	0.762 [0.720, 0.805]	0.069 [0.054, 0.100]
Full	0.032 [−0.045, 0.086]	0.782 [0.736, 0.828]	0.080 [0.058, 0.115]
AUC-PR lift, paired			1.3× [1.0, 2.1]
AUC-PR lift (Full), paired			1.5× [1.0, 2.5]

Table 4: Out-of-sample onset prediction under leave-one-country-out cross-validation ($N = 8,835$ country-years, 184 onsets; AUC-PR no-skill baseline = 0.021). *Entrants* adds the two aggregate shadow variables to the *Baseline*; *Full* resolves the shadow by intervener type (Section 3.4). Brackets hold 95% intervals from a country-cluster bootstrap of the held-out predictions ($B = 1,000$); the lift rows report the AUC-PR ratio over *Baseline*, paired within each resample.

Specification	PRL	ROC-AUC	AUC-PR
<i>Cunningham subsample</i> ($N = 6,932$)			
Baseline	0.083	0.752	0.062
+ Cunningham	0.081	0.750	0.062
Entrants	0.117	0.783	0.082
+ Cunningham	0.115	0.781	0.081
Full	0.204	0.861	0.151
+ Cunningham	0.202	0.860	0.148
<i>G&M subsample</i> ($N = 7,875$)			
Baseline	0.080	0.746	0.063
+ G&M	0.072	0.734	0.061
Entrants	0.116	0.782	0.084
+ G&M	0.107	0.774	0.080
Full	0.206	0.861	0.156
+ G&M	0.200	0.859	0.142

Table 5: Out-of-sample metrics (leave-one-country-out), computed on each proxy’s coverage subsample. Each panel adds the Cunningham security-hierarchy measure or the Gibilisco–Montero (G&M) structural P5 estimates to the *Baseline*, *Entrants*, and type-resolved *Full* models.

	Coef.	$P(\text{onset})$, 10th $\rightarrow$ 90th representative country	Sign-consistent of 25 draws
Net tilt ($E^O - E^G$) [−1.34, +3.79]	+0.90	0.70% $\rightarrow$ 1.17%	84%
Common intensity ($E^O + E^G$) [−2.32, −0.18]	−0.71	1.94% $\rightarrow$ 0.22%	100%

Table 6: Direction of the shadow’s effect on onset. The collinear raw channels ($\gamma^G = -1.61$, $\gamma^O = +0.19$; $r \approx 0.94$) reparameterize into a net tilt ($E^O - E^G$) and a common intensity ($E^O + E^G$). Intervals pool the $T \times P$ bootstrap—measurement spread and sampling together. $\Delta P(\text{onset})$ is the change at a mean-covariate country as the axis moves from its tenth to its ninetieth percentile. Full model, Table 15.

Shadow channel	Aggregate	By type
Government	0.07	0.10
Opposition	0.06	0.10
Both	0.07	0.14

Table 7: Out-of-sample onset prediction (AUC-PR, leave-one-country-out). Each cell is a separate onset model: the structural baseline plus the indicated channel, entered as a single aggregate expectation or split into the six intervener types.

Intervener type	OOS AUC-PR	Government side	Opposition side
Neighbors	0.09	deters (100%)	emboldens (92%)
Powers	0.09	deters (80%)	deters (88%)
Rivals	0.09	deters (100%)	emboldens (92%)

Table 8: Direction of the shadow’s effect, by intervener type. Each entry gives the majority direction and its consistency across the 25 measurement draws.

Online Appendix

A The Onset Model

This appendix supplies the formal model behind Section 1. Parts A.1 and A.2 build the onset game of Figure 1 and derive Equation (1), the object the two empirical stages estimate. Parts A.3 and A.4 give the war payoff its structure and embed it in the bargain. Parts A.5–A.7 work the bargain once under each of the three common frictions; Part A.8 states Expectations 1 and 2, the comparative statics Section 3 tests; and Part A.9 derives the per-type weights read by its heterogeneity analysis.

Notation. O and G are the opposition and the government; $i = 1, \dots, n$ indexes the potential interveners, each choosing $a_i \in \{O, N, G\}$ —back the opposition, stand aside, back the government; σ_i collects intervener i ’s probabilities $(\sigma_i(O), \sigma_i(G))$; $E^O = \sum_i \sigma_i(O)$ and $E^G = \sum_i \sigma_i(G)$ are the anticipated opposition- and government-biased intervention a country faces; $\tau = E^O - E^G$ is the net tilt and $\iota = E^O + E^G$ the common intensity; $\pi(\tau)$ is the contest probability and $k_O(\iota)$, $k_G(\iota)$ the war burdens, with $k \equiv k_O$ where unambiguous; $\bar{x}$ is the credible-concession ceiling (Part A.5); q , κ , and $\delta(\iota)$ are the screen’s prior, resolve gap, and intensity increment (Part A.6); τ_O , τ_G , and ε are the two reads of the tilt and the optimism parameter (Part A.7); and w_i and b_i are type-specific contest and burden loadings (Part A.9).

A.1 The game and its shadow

The decision problem of Figure 1 in the main text has three moves. First, the opposition O chooses whether to mount a challenge; if it does not, the status quo prevails. Second, a challenge opens a bargain over control of the state—a stake normalized to 1: O presses a demand $s \in [0, 1]$, which the government G accepts—a settlement, on those terms—or rejects, in which case war follows. The bargain is worked in Parts A.4–A.7; for the game form, what matters is that it ends in one of two ways, a settlement or a war. Third, in war, the n potential interveners respond simultaneously—each backing the opposition, the government, or neither—and their combined response is the international environment in which the war is fought.

It is this third move that casts the shadow. Because O and G bargain in

anticipation of how outside powers would respond, the interveners’ equilibrium response—call it σ^* —can decide between peace and war before a shot is fired (Cetinyan, 2002). Intervenors’ payoffs depend on the war and on one another’s actions, not on the demand whose rejection produced it, so the intervener subgame is the same at every war node and its equilibrium σ^* is a single, well-defined object. Write $U_O(F)$ for the opposition’s expected payoff from a war so shaped: it is the disagreement point of the bargain and the object the rest of this appendix works with.

Remark (rational expectations). Whether σ^* approximates the expectations that rebels actually form is not merely assumed. Under rational expectations, agents exploit available information at least as well as the researcher’s model (Muth, 1961). The Stage 1 classifier described in Section 2 recovers a substantial share of the predictive content in realized intervention outcomes using only publicly observable state characteristics; rebels who can also draw on private intelligence, direct patron relationships, and local knowledge—and who stake their lives on the answer—face a forecasting problem that is easier, not harder, than the one the classifier solves.

A.2 Aggregation

The shadow σ^* is a Nash equilibrium of the intervention subgame: each potential intervener i has a utility over the profile $a = (a_1, \dots, a_n) \in \{O, N, G\}^n$ of all interveners’ actions, and at σ^* each plays a best response to the rest. Stated exactly, the opposition’s problem is intractable: with every other state in the panel a potential intervener—as many as 160 in recent decades—the expected war payoff runs over the 3^n profiles of their joint play, more than the number of elementary particles in the observable universe. Two assumptions reduce the problem to a tractable form without discarding its content.

Assumption 1 (Separability). The opposition’s payoff from any profile is additive across interveners: $u_O(a) = \sum_i u_O^i(a_i)$, where $u_O^i : \{O, N, G\} \rightarrow \mathbb{R}$ measures the marginal contribution of intervener i ’s action.

Separability captures a specific and substantively defensible theory of how opposition groups aggregate international expectations: each outside power’s likely behavior enters the calculus independently, and the combined effect is their sum. A rebel coalition assessing its international environment in 1970s Angola had a view about what Cuba would likely do, a separate view about South Africa, and a separate view about the United States; the hypothesis is that these entered additively. The assumption rules out complementarities—

the value of Soviet logistics may depend on whether Cuban troops are also present—but coordinated multilateral operations constitute a small minority of actual interventions.⁷

Assumption 2 (Non-intervention as reference). For all i , $u_O^i(N) = 0$. This normalization makes the intervention-free onset model a limiting case: if no state intervenes, the expected utility of fighting reduces to purely domestic considerations, and the standard country-year onset model follows directly.

Under these two assumptions the opposition’s expected payoff from war collapses to a sum of marginal terms. For any joint distribution over profiles with marginal intervention probabilities σ_i , linearity of expectation gives $\mathbb{E} u_O(a) = \sum_i \mathbb{E} u_O^i(a_i)$, and the standing-aside terms vanish by Assumption 2, so that

$$U_O(F; \sigma) = \sum_{i=1}^n [\sigma_i(O) u_O^i(O) + \sigma_i(G) u_O^i(G)] \quad (1)$$

—Equation (1) of the main text, here derived rather than posited. The marginal probabilities suffice however correlated the interveners’ actions are.

That sufficiency is what carries the model to the full system of potential interveners. Computing the opposition’s expected utility requires only the marginals σ_i , not the joint distribution over all n players’ actions simultaneously. Specifying and estimating the full joint game—the approach of Gibilisco and Montero (2022) for the five permanent Security Council members—becomes computationally intractable beyond a small player set. Separability sidesteps that constraint while preserving the game’s central implication: the equilibrium strategies of outside powers must be mutually consistent. That consistency requirement is enforced directly via the Nash fixed-point iteration described in Section 2.

Equation (1) has two kinds of unknowns—the probabilities σ_i , estimated in Section 2, and the weights u_O^i , recovered from onset behavior in Section 3—

⁷Aydin and Regan (2012) find that the *alignment* among interveners matters for civil war duration: same-side interventions shorten wars only when the intervening states share similar preferences. This is evidence of complementarities in the conduct phase. Separability here pertains to the onset calculus—the rebel’s assessment of the intervention environment at the moment it decides whether to fight—not to the downstream interactions among interveners once fighting begins. The spatial lags in the Stage 1 classifier (Section 2) capture interdependence among interveners’ *choices* without requiring that their *effects* on the rebel enter non-additively.

and the object of interest is the sum at the intervener equilibrium, $U_O(F; \sigma^*)$. The next part gives the weights their structure.

A.3 The war payoff

Two forces run through the weights u_O^i , and they organize everything that follows. The first is the *contest*: support aimed at a side improves that side's odds of prevailing. The government-side half of this is exactly the deterrent logic of [Cunningham \(2016\)](#)—the prospect of outside backing for the incumbent lowers the rebels' expected return to fighting—stated here for the full population of potential sponsors rather than the single great-power patron on which that analysis centers. The second is the *burden*: whichever side an entrant would favor, its involvement makes the war it enters a bigger war—longer, heavier, harder on the parties who must fight it. An anticipated intervener therefore cuts twice: once through whom the war would favor, and once through what the war would cost. The payoff that carries these forces is the standard one—the opposition prevails with some probability and bears some cost, [Fearon's](#) p and c —and the shadow enters by tying each primitive to one axis: the odds to the tilt, the burden to the intensity.

Assumption 3 (Contest and burden). With $\tau \equiv E^O - E^G$ (the net tilt) and $\iota \equiv E^O + E^G$ (the common intensity), the war payoffs, in shares of the stake, are

$$U_O(F) = \pi(\tau) - k_O(\iota), \quad U_G(F) = (1 - \pi(\tau)) - k_G(\iota),$$

where the contest probability π is increasing in the tilt, each burden k_O , k_G is increasing in the intensity, and all three are affine over the observed range; states expected to stand aside add no burden (the normalization of Assumption 2). Where only the opposition's burden is in play, write $k \equiv k_O$ and $k' \equiv k'_O$.

Assumption 3 states the war payoff in aggregate coordinates; Equation (1) requires it to be additive across interveners with fixed marginal weights. Affinity is what reconciles the two. Because π and k_O are affine, each can be written exactly as its intercept plus a constant slope, $\pi(\tau) = \pi(0) + \pi'\tau$ and $k_O(\iota) = k_O(0) + k'\iota$. Substitute the definitions $\tau = E^O - E^G$ and $\iota = E^O + E^G$,

with $E^O = \sum_i \sigma_i(O)$ and $E^G = \sum_i \sigma_i(G)$:

$$\begin{aligned} U_O(F) &= \pi(\tau) - k_O(\iota) \\ &= [\pi(0) - k_O(0)] + \pi' (E^O - E^G) - k' (E^O + E^G) \\ &= [\pi(0) - k_O(0)] + \sum_{i=1}^n \left[(\pi' - k') \sigma_i(O) - (\pi' + k') \sigma_i(G) \right]. \end{aligned}$$

The sum is Equation (1), with the same pair of marginal weights for every intervener,

$$u_O^i(O) = \pi' - k', \quad u_O^i(G) = -(\pi' + k');$$

a curved π or k would instead make intervener i 's marginal contribution depend on the rest of the environment, breaking the additive form. The bracketed intercept $\pi(0) - k_O(0)$ is the war value absent any anticipated intervention—the purely domestic term of Assumption 2's reference point—constant in the shadow and absorbed empirically into the second stage's domestic baseline $\mathbf{X}\beta$; Equation (1) records the shadow-dependent part, through which every derivative below runs. Two readings follow at once. On the government side the two forces agree: backing for the incumbent worsens the rebels' odds *and* heightens the war's burden, so the theory signs that channel unambiguously, $-(\pi' + k') < 0$. On the opposition side they compete: outside backing improves the rebels' odds but heightens the burden of the very war in which the help would arrive, and $\pi' - k'$ is a race the theory does not resolve. The natural coordinates of the war payoff are therefore not the two channels but the two axes—the tilt, which moves the contest, and the intensity, which moves the burden—the combinations the second-stage analysis can estimate with any stability (Section 3).

Remark (why affine). Affinity is a local claim, and four observations bound its cost. The observed tilt spans a short arc—the second stage's tenth-to-ninetieth-percentile move covers well under a unit on the asinh scale—over which any smooth contest function is approximated by its chord. The tilt is a difference of matched, co-moving aggregates, so it operates in the interior of any S-shaped contest function, where curvature is second-order. The estimation scale is already concave in raw counts—the channels enter through asinh—so affinity in τ still embodies diminishing returns to additional interveners. Affinity is also exactly mean-shadow sufficiency, $\mathbb{E}[\pi(\tau)] = \pi(\mathbb{E}\tau)$: without it the shadow's variance, not just its level, would become payoff-relevant, and the measure would owe the reader second moments. And the

conclusions do not hang on it: the next Remark shows the channel signs survive on monotonicity alone, and the interaction probe of Part A.5 turns up none of the negative tilt-by-intensity product a scaling burden would imply.

Remark (general form). Nothing below turns on the exact separation of contest from burden in Assumption 3. With a general $\pi(E^O, E^G)$, increasing in E^O and decreasing in E^G , and a general $k(E^O, E^G)$, weakly increasing in both, the channel-coordinate conclusions hold unchanged— $\partial U_O(F)/\partial E^G = \pi_{EG} - k_{EG} < 0$ always, while $\partial U_O(F)/\partial E^O = \pi_{EO} - k_{EO}$ is unsigned—and the axis conclusions hold iff direction moves the contest more than the burden, $\pi_{EO} - \pi_{EG} > k_{EO} - k_{EG}$, and intensity moves the burden more than the contest, $k_{EO} + k_{EG} > \pi_{EO} + \pi_{EG}$; both are automatic under the separated form. The empirical mapping is a rotation, not a test: the second stage’s axis coefficients estimate $(\pi', -k')$ directly. Two overidentifying inequalities do follow for the raw channels—the government-side effect exceeds the opposition-side effect in magnitude, $\pi' + k' > |\pi' - k'|$, and it exceeds the intensity effect, $-(\pi' + k') < -k' < 0$ —and both hold in the estimates of Section 3.

A.4 The bargaining stage

Now embed the war value in the bargain. The protocol is the game’s second move: on challenging, O presses its demand $s \in [0, 1]$ (its share; G keeps $1 - s$), which G accepts—ending in peace—or rejects, in which case war follows. G accepts iff $1 - s \geq U_G(F)$, that is, iff

$$s \leq \pi(\tau) + k_G(\iota). \quad (3)$$

Absent any friction the bargain never breaks. The largest demand G accepts, $s = \pi + k_G$, exceeds O ’s war payoff $\pi - k_O$ by the joint burden $k_O + k_G > 0$, so O demands exactly that, G accepts, and war never occurs—neither axis moves onset, because onset never varies. The three parts that follow restore one friction at a time—a commitment ceiling, private resolve, divergent reads of the shadow—each in the vocabulary of Assumption 3.

A.5 The commitment cutpoint

Suppose first that the war costs are common knowledge but that G cannot credibly transfer more than a share $\bar{x} < 1$ of the stake: it cannot bind

its later self to honor a larger concession once the external threat recedes. Feasible peaceful offers are then those with $s \leq \bar{x}$. Since O accepts no share below its war payoff $U_O(F)$, a peaceful division exists iff $U_O(F) \leq \bar{x}$, and war—onset—occurs exactly when

$$U_O(F) = \pi(\tau) - k_O(\iota) > \bar{x}. \quad (4)$$

(When the ceiling blocks peace it is the ceiling, not G 's willingness, that blocks it: $U_O(F) \leq \pi + k_G$ always, so a demand G would accept exists but exceeds what G can credibly deliver.) Because π rises in the tilt and k_O in the intensity, (4) is a cutpoint: holding the rest fixed, onset switches on as the tilt rises and off as the intensity rises. Across a population of country-years with heterogeneous $(\bar{x}, k_O)$, the probability of onset is $\Pr(\pi(\tau) - k_O(\iota) > \bar{x})$ —a smooth function increasing in the tilt, decreasing in the intensity, and saturating in $\pi \in [0, 1]$, the form the second-stage link estimates, with unobserved heterogeneity in the threshold as its error (the domestic intercept $\pi(0) - k_O(0)$ and the threshold's observable movers are the second stage's $\mathbf{X}\beta$).⁸

The tilt relationship is weakly monotone, never sign-reversing, provided the credible ceiling does not rise with the tilt faster than the contest does: $\partial \bar{x} / \partial \tau < \pi'$ suffices for single crossing of (4) in τ . This is precisely where the account parts from [Langø \(2023\)](#), who shows that anticipated opposition-biased intervention can *lower* onset: by tipping the balance of power against the government, it compels some governments to acquiesce—to concede power outright—rather than fight. That pacifying channel runs through *credible* concession, modeled there as a terminal transfer of power; the commitment friction is precisely its denial—the assumption that credible concession is bounded and does not grow with the opposition's external backing. The two accounts agree on what is at issue and part only there. We adopt the friction, and the data lean its way: the tilt of the anticipated environment toward the opposition predicts *more* onset, not less (Section 3).⁹

⁸The additive form of (4) carries a testable implication: a burden that *scaled* the contest rather than adding to it would sign a tilt-by-intensity product term negative on the index scale. Re-estimating the second-stage model with that product term returns no such negative sign—the coefficient is instead weakly positive (positive in 76 percent of the 25 measurement draws), so the scaling variant is disfavored. The interaction is not sharply estimated in either direction, as one expects of a product of two channels that are themselves nearly collinear, and none of the paper's claims depends on its form.

⁹The qualitative record agrees the net pull is toward war, with rebel leaderships doc-

A.6 The common-prior resolve screen

Now drop the ceiling and restore the standard information friction: G 's resolve is private. The bargain that results is the ubiquitous ultimatum screen (e.g., [Slantchev and Tarar, 2011](#)), transposed to the onset setting. G is resolute or soft: the resolute type bears only the common intensity burden $\delta = \delta(\iota)$ that war imposes on every combatant (O 's war cost is $k_O + \delta$; the setup's $k_O(\iota)$ thus decomposes as an intensity-free base plus the common increment), while the soft type bears an additional resolve gap $\kappa > 0$; $q \equiv \Pr(\text{resolute})$ is common knowledge. By (3) the resolute type accepts any demand $s \leq \pi + \delta$ and the soft type any $s \leq \pi + \kappa + \delta$, so the soft type accepts more. Two pure demands are undominated. The *safe* demand $s = \pi + \delta$ is accepted by both types and yields O that amount with certainty. The *risky* demand $s = \pi + \kappa + \delta$ is accepted only by the soft type: with probability $1 - q$ it yields $\pi + \kappa + \delta$, and with probability q it is refused and war yields $U_O(F) = \pi - k_O - \delta$. O prefers the risky demand iff

$$(1 - q)(\pi + \kappa + \delta) + q(\pi - k_O - \delta) \geq \pi + \delta,$$

which reduces to

$$(1 - q)\kappa \geq q(k_O + 2\delta). \quad (5)$$

Two things have happened. The balance of power $\pi(\tau)$ has dropped out: it shifts both acceptance thresholds and O 's war payoff by the same amount, relocating the division but cancelling from the risky-versus-safe comparison. The burden has not: $\delta(\iota)$ survives on the right of (5)—the war O risks is costlier, while the safe demand, tracking both types' diminished war values, extracts more—so the equilibrium probability of war,

$$\Pr(\text{onset}) = q \cdot \mathbf{1}[(1 - q)\kappa \geq q(k_O + 2\delta)],$$

is weakly *decreasing* in the intensity and independent of the tilt. At the corner, where π is high enough that demands are capped at the whole stake, raising π only *lowers* the chance of war, as O comes to extract the entire stake peacefully; the screen therefore never produces a positive tilt effect, interior or corner.

umented conditioning the decision to fight on anticipated outside support ([Kuperman, 2008](#); [Kydd and Straus, 2013](#)).

Two features of the cost structure are load-bearing. The intensity increment is *additive* and *type-common*: it attaches to the character of the war—longer, heavier, more internationalized—rather than to the government’s resolve. A multiplicative burden scaling all costs equally would cancel from the interior comparison, and an increment that fell disproportionately on the soft type could reverse the sign; the substantive content of Assumption 3’s burden is exactly the type-common reading.

Remark (screening on capacity). Nothing here turns on resolve being the private object. Let the government’s type instead be its capacity: O prevails with probability $\pi_L(\tau)$ against a strong government and $\pi_H(\tau)$ against a weak one, $\pi_H(0) > \pi_L(0)$, both types shifted equally by the tilt, with the war costs $k_O + \delta$ and $k_G + \delta$ common knowledge. The screen replays: the weak type accepts more, the safe demand is $\pi_L + k_G + \delta$, and the risky demand $\pi_H + k_G + \delta$ is refused by the strong type—against whom the war is then fought. The risky demand is preferred iff

$$(1 - q) \Delta\pi \geq q(k_O + k_G + 2\delta), \quad \Delta\pi \equiv \pi_H(0) - \pi_L(0),$$

with q now the probability of the strong type. The tilt enters the safe demand, the risky demand, and the war payoff alike, and cancels with weights $(1 - q) + q - 1 = 0$; the intensity survives in 2δ . The row is unchanged: what is screened supplies the spread—the resolve gap κ or the capacity gap $\Delta\pi$ —but the cancellation belongs to the screen itself, so a positive tilt effect excludes it whatever the private object. (As with costs, type-dependence in the *sensitivity* to the tilt, $\pi'_H > \pi'_L$, would let the spread grow with τ and transmit it; the parallel caveat applies.)

The screen, then, transmits the intensity and blocks the tilt: onset falls in ι here as under commitment, but no positive tilt effect can arise through this channel. It is the tilt’s presence in the data, not the intensity’s, that discriminates among the frictions (Expectation 2, Part [A.8](#)).

A.7 Divergent expectations

Finally, drop the private resolve and let the parties read the shadow itself differently. The shadow is a forecast of third parties’ conduct in a war that has not happened; no pre-war exchange can realize it, so the government and the rebels can hold—and know they hold—different estimates of it ([Cetinyan, 2002](#); [Thyne, 2006](#)). Held as common knowledge, such divergence departs

from common priors (Aumann, 1976): among common-prior rationals, different reads are private information, and the bargain processes private information as the screen (cf. Slantchev and Tarar, 2011). This friction therefore runs on priors that differ—a primitive the other two bargains do not need. Let τ_O and τ_G be the two reads of the tilt, common knowledge, with war costs $k_O(\iota)$ and $k_G(\iota)$ as above. G accepts any demand $s \leq \pi(\tau_G) + k_G$, and O , knowing this, compares the best acceptable settlement with war as *it* sees war: onset occurs iff $\pi(\tau_O) - k_O > \pi(\tau_G) + k_G$, that is, iff

$$\pi(\tau_O) - \pi(\tau_G) > k_O(\iota) + k_G(\iota). \quad (6)$$

War requires the rebels’ optimism about the tilt to exceed the joint burden of fighting. Under proportional optimism— $\tau_O = (1 + \varepsilon)\tau$, $\tau_G = \tau$, $\varepsilon > 0$ —the wedge is $\varepsilon\pi'\tau$, growing wherever the environment tilts toward the opposition; across country-years heterogeneous in $(\varepsilon, k_O + k_G)$, the probability of onset rises in τ (on $\tau > 0$) and falls in ι . Two limits mark the channel’s shape: it is one-sided, silent where the environment tilts against the opposition, and it saturates, the wedge vanishing as π approaches its bounds. Under a common prior the construction would unravel: because war here requires both a demand the government will refuse and the refusal itself, each side must read the other’s willingness to fight as information, and Fey and Ramsay (2007) show that no equilibrium war survives those inferences—war by mutual optimism requires beliefs a common prior cannot deliver. The channel therefore rests on heterogeneous priors, and what makes that escape unusually defensible here is the object of the disagreement: the counterfactual conduct of third parties, which no pre-war exchange between the combatants can arbitrate and only war itself reveals. This is the mechanism of Cetinyan (2002) and Thyne (2006) in the present notation: not the potential for intervention but different expectations of it is what moves onset through this channel.

A.8 What the frictions transmit

The three instances work the bargain once under each friction, all in the vocabulary of Assumption 3, and their comparison yields a taxonomy: the two axes carry different evidentiary weight. That onset falls with the intensity of the anticipated environment is close to common property—any costly-war bargain that produces war with positive probability delivers it. That onset rises with the tilt is the diagnostic fact: no frictionless or common-prior

account transmits it, so its presence in the data is a trace of a friction that lets the balance of power reach the decision to fight.

Expectation 1 (Axes of the shadow). Under the commitment friction, the probability of onset is increasing in the net tilt $\tau = E^O - E^G$ and decreasing in the common intensity $\iota = E^O + E^G$ of anticipated intervention (by Equation (4) and the weights of Part A.3). In channel coordinates the government-side effect is $-(\pi' + k') < 0$ —contest and burden push together—while the opposition-side effect is the race $\pi' - k'$, which the theory does not sign. The account is falsified by a robustly negative tilt effect or a robustly positive intensity effect; it is falsified by neither sign of the raw opposition channel, whose fragility it expects.

Expectation 2 (What the frictions transmit). The probability of onset falls in the common intensity under all three bargains—weakly under the common-prior screen, strictly under the commitment ceiling and divergent expectations—but rises in the net tilt only under the latter two: under commitment throughout, under divergent expectations one-sidedly, on the region where the environment already tilts toward the opposition. Under the common-prior screen it cannot rise in the tilt, interior or corner. A positive tilt effect therefore excludes the frictionless and common-prior accounts, while leaving commitment and divergent expectations in contention; a negative intensity effect is common property, diagnostic of the burden rather than of any friction.

Bargain	Effect on $\Pr(\text{onset})$ of		Derived at
	tilt τ	intensity ι	
Frictionless	none: war never occurs		Part A.4
Common-prior screen	0 interior; ≤ 0 at corner	– (weakly)	(5)
Commitment ceiling	+	–	(4)
Divergent expectations	+ on $\tau > 0$; silent on $\tau \leq 0$	–	(6)

The exclusion is of the screen as written. The load-bearing caveats of Part A.6 mark the escape routes—a burden that scales rather than adds, an increment that falls disproportionately on the soft type, type-dependent sensitivity to the tilt—and a screening variant built along any of them is a legitimate rival, testable against data richer than these.

Onset data cannot separate the two surviving frictions, and we do not ask them to. We lead with commitment for its prominence across the civil-

war record, its aptness to military intervention, where the calculus of victory really is altered (Walter, 2009), and its economy of primitives: the ceiling’s source is ready-made—settle, the external threat recedes, and with it whatever concession it extracted (Part A.5)—while divergence runs on the differing priors Part A.7 must posit (Aumann, 1976). The divergent-expectations reading remains a live alternative rather than a rival.

A.9 Heterogeneous interveners

One more consequence of Equation (1) is worth setting down, because Section 3 uses it. Nothing in Assumption 1 requires the weights to be uniform across interveners; uniformity entered with Assumption 3’s unweighted aggregates. Generalize them: let $w_i \geq 0$ and $b_i \geq 0$ be type-specific contest and burden loadings—how much intervener i ’s entry would move the balance, and how much it would internationalize the war—and let

$$U_O(F) = \pi(\tau_w) - k(\iota_b),$$

$$\tau_w \equiv \sum_i w_i [\sigma_i(O) - \sigma_i(G)], \quad \iota_b \equiv \sum_i b_i [\sigma_i(O) + \sigma_i(G)].$$

The weighted aggregator is the assumption; the per-type weights are its arithmetic. Expanding as before, $u_O^i(O) = \pi'w_i - k'b_i$ and $u_O^i(G) = -(\pi'w_i + k'b_i)$, with Assumption 3 the case $w_i = b_i = 1$. The government-side weight is non-positive for every type, strictly negative wherever either loading is positive: no anticipated intervener emboldens rebellion from the government’s side. The opposition-side race, by contrast, may resolve differently for different interveners: a state whose involvement would internationalize the war more than tilt it ($k'b_i > \pi'w_i$) deters even from the opposition’s side—the signature Section 3 reads in the major-power reversal. The theory fixes the aggregate axes and the government channel; which interveners carry which weight is a question for measurement.

Remark (what the model stakes). Three limits are built into the empirical exercise this model serves. The shadow is estimated, not observed, so effect magnitudes arrive blurred by measurement error; its two channels co-move tightly, so only their sum and difference are stable; and the observable is onset, not the terms of the settlements that avert it, so the friction is detectable only through its trace. The model is pitched at the altitude of signs deliberately—strong enough that a reversed sign would falsify it, no stronger

than the measurement can carry. And no claim is made that any one of these games is *the* mechanism: the pair of comparative statics is what any account with a monotone contest, an intensity-borne burden, and either surviving friction implies.

B Regan Intervention Coding Corrections

The raw [Regan \(2000\)](#) data contain ten per-conflict coding decisions that are corrected before analysis. These are listed in Table 9. In all cases the corrections are applied to the raw Stata file *before* country-code standardization, matching the original R pipeline’s order of operations.

After corrections, two directed dyads with conflicting target assignments across multiple records are resolved by fixing the target directly: Georgia–Russia (opposition-biased) and Liberia–Sierra Leone (government-biased). Final deduplication retains the first record per directed dyad-year after sorting by the `ddyear` identifier.

B.1 Post-1999 Intervention Coding

The [Regan \(2000\)](#) data end in 1999. To extend the Stage 1 training set through 2014, I code military interventions in all 38 post-1999 COW onset country-years using the same threshold: deployment of military personnel (troops, combat advisors, or directed proxy forces) attributable to a specific state, on behalf of one side. Arms transfers, economic aid, sanctuary, and strategic direction alone are insufficient. UN peacekeeping missions are excluded. Table 10 lists the 32 coded interventions; the remaining 17 onset country-years are coded as having no intervention meeting the threshold. Full source notes for each coding decision are available in the replication archive.

Onset country-years coded as having no military intervention meeting the personnel-deployment threshold: Guinea (2000), Philippines (2000, 2003, 2005), Burundi (2001), Rwanda (2001), Nepal (2001, 2003), Sudan (2003, 2011), Indonesia (2003), Yemen (2004, 2007), Sri Lanka (2006), Syria (2011), Myanmar (2011), and Central African Republic (2012).

Conflict	State	Correction	Rationale
Congo-Brazzaville (1997)	France (220)	target: neutral $\rightarrow$ 2 (opp)	France supplied arms to Cobra militia, siding with Angola/opposition
Indonesia (1958)	USA (002)	target: gov $\rightarrow$ 2 (opp)	CIA Operation Archipelago backed anti-communist rebels
Cameroon (conflict 17)	W. Germany	ccode: 255 $\rightarrow$ 260	Miscoded; correct COW code for West Germany
Congo Crisis (1960s)	All	host ccode: 484 $\rightarrow$ 490	DRC better represents this conflict than Republic of Congo
Congo Crisis (1964)	Belgium (211)	target flipped gov $\leftrightarrow$ opp	Sides effectively reversed in 1964 phase
Ethiopia (1980s)	Somalia (520)	target: gov $\rightarrow$ 2 (opp)	Somalia generally supported Ogaden opposition
Cambodia	Vietnam (816)	target: opp $\rightarrow$ 1 (gov)	Vietnam supported incumbent government throughout
Mozambique	Zimbabwe (552)	target: $\rightarrow$ 1 (gov)	Zimbabwe consistently backed FRELIMO government
Mozambique	Malawi (553)	target: $\rightarrow$ 3 (neutral)	Malawi supported both sides; excluded as neutral
Zimbabwe/South Africa	South Africa	target: $\rightarrow$ 3 (neutral)	Supported both sides; excluded
Sri Lanka	UK (200)	target: $\rightarrow$ 2 (opp)	Miscoded; correct side assignment
Sri Lanka	India (750)	target: $\rightarrow$ 1 (gov)	India supported government
Iran	Iraq (645)	target: gov $\rightarrow$ 1	Iraq favored government; miscoded in original

Table 9: Regan coding corrections applied before analysis.

C Multiple Imputation Procedure

Missing values in continuous predictors are handled through a two-stage multiple imputation strategy using miceforest (Multiple Imputation by Chained Equations with LightGBM). The strategy generates $5 \times 5 = 25$ complete datasets; estimates are combined across datasets following [Rubin \(1987\)](#).

The staged structure reflects the data’s own hierarchy and is a design choice, not a computational convenience. Country-year quantities (GDP per capita, regime scores, CINC components) must be imputed at the country-year level: imputing them at the directed-dyad level would treat state A ’s

Host	Year	Intervener	Side
Sierra Leone	2000	United Kingdom	Gov.
Ivory Coast	2002	France	Gov.
Liberia	2002	Guinea	Opp.
Pakistan	2004	United States	Gov.
Chad	2005	France	Gov.
Chad	2005	Sudan	Opp.
Pakistan	2005	United States	Gov.
Somalia	2006	Ethiopia	Gov.
Chad	2007	France	Gov.
Chad	2007	Sudan	Opp.
Pakistan	2007	United States	Gov.
Somalia	2009	Ethiopia	Gov.
Iraq	2010	United States	Gov.
Ivory Coast	2011	France	Opp.
Libya	2011	United States	Opp.
Libya	2011	United Kingdom	Opp.
Libya	2011	France	Opp.
Libya	2011	Qatar	Opp.
Libya	2011	UAE	Opp.
Mali	2012	France	Gov.
Mali	2012	Chad	Gov.
Nigeria	2013	Chad	Gov.
South Sudan	2013	Uganda	Gov.
Ukraine	2014	Russia	Opp.
Somalia	2014	Ethiopia	Gov.
Somalia	2014	Uganda	Gov.
Libya	2014	Egypt	Gov.
Libya	2014	UAE	Gov.
Yemen	2014	Saudi Arabia	Gov.
Yemen	2014	UAE	Gov.
Afghanistan	2014	United States	Gov.
Afghanistan	2014	United Kingdom	Gov.

Table 10: Post-1999 military interventions in COW onset country-years.
Gov. = government-biased; Opp. = opposition-biased.

GDP as a separate quantity in each of A 's dyads, producing inconsistent imputed values across rows that share the same country and year. Dyadic quantities (bilateral trade, peace scores, capital distances) must be imputed before directed expansion so that the symmetry constraint $x_{AB} = x_{BA}$ holds within each imputed dataset—a constraint that would be broken if each directed-dyad row were imputed independently. Imputing in sequence, country-year first and undirected dyad second, preserves both requirements simultaneously.

Stage 1—Country-year imputation. The analysis sample (1946–2014, $n = 8,897$ country-years) is imputed five times. Variables imputed include regime scores, GDP per capita, population, oil wealth, mountainous terrain, ethnic fractionalization, ethnic exclusion, all six NMC capability components, and UN ideal points. Six right-skewed NMC variables are asinh-transformed before imputation (with individually optimised scale parameters minimising the Kolmogorov–Smirnov statistic against the standard normal) and back-transformed afterwards. Five iterations of chained equations are run. Post-imputation clipping enforces substantive constraints ($-10 \leq \text{polity2} \leq 10$, probabilities $\in [0, 1]$, capabilities ≥ 0).

Stage 2—Undirected-dyad imputation. For each of the five country-year datasets, the undirected dyad frame ($\sim 599,000$ rows) is imputed five times before expansion to directed dyads. Imputation at the undirected-dyad stage ensures that symmetric quantities (trade flows, peace scores, capital distance) satisfy $x_{AB} = x_{BA}$ by construction within each imputed dataset. Variables imputed include bilateral trade flows, peace quality scores, capital distances, and dispute outcome expectations. Two iterations of chained equations are run.

The 25 resulting complete datasets are processed identically through the spatial-weights and intervention-coding stages before combination.¹⁰

¹⁰The 25 branches are mutually independent at every stage upstream of the final combination: each (cy, ud) pair flows from imputation through Stage 1 training, Stage 1 prediction, and Stage 2 estimation without reference to any other branch. Rubin's Rules averaging is the only step that requires all 25 to be complete. This means the pipeline is embarrassingly parallel—all 25 classifiers could in principle be trained simultaneously—and that updating the sample in a future study would not require retraining all 25 models from scratch. A single additional imputed dataset, processed through the same pipeline, contributes marginally to the final combination.

D Variable List

Feature sources: material capabilities from COW v6 (Singer, 1987) and V-Dem v15 (V-Dem Institute, 2025); ethnic structure from Fearon and Laitin (2003) and Cederman et al. (2010); alliances (Gibler, 2009); contiguity (Stinnett et al., 2002); territorial disputes (Huth and Allee, 2003); expected dispute outcomes (Carroll and Kenkel, 2019); peace quality (Diehl et al., 2021); UN General Assembly ideal points (Bailey et al., 2017); and IGO membership (Pevehouse et al., 2020).

Table 11 lists all country-year variables included in the analysis. Table 12 lists the dyadic variables added at the directed-dyad stage.

E Spatial Weights Derivation

For each year t in the analysis window, I construct a row-normalized polity-similarity weight matrix W_t over the set of potential intervener states. Let p_i denote the Polity II score of state i in year t . The raw weight is:

$$w_{ij} = \begin{cases} 1/|p_i - p_j| & \text{if } p_i \neq p_j \text{ and } i \neq j, \\ 0 & \text{otherwise.} \end{cases}$$

Row-normalization gives $W_{ij} = w_{ij}/(\sum_k w_{ik})$, so that each row sums to 1 (rows summing to zero remain zero). States with missing Polity scores are excluded from W_t in that year.

The polity-similarity weighting encodes the assumption that states pay more attention to the intervention choices of ideologically similar peers, and less to the choices of ideological adversaries. This departs from the standard inverse-distance spatial weight used in most spatial-lag models, which would give *higher* weight to more *different* states; the similarity-based weight is theoretically more appropriate for the ideological-diffusion channel.

Yemen (COW code 678) is excluded from the 1990 weight matrix to avoid anomalous results during the unification transition year.

For each onset event (host country A , year t), the spatial lag for potential intervener B is the weighted sum of other potential interveners' predicted intervention probabilities, with A removed from the weight matrix before computation:

Variable	Description	Source	Coverage
onset	Civil war onset (COW types 4–5, $\geq$ 1,000 battle deaths)	COW ISW v5.1	1946–2014
polity2	Polity II score (−10 to +10)	V-Dem v15 (e.polity2)	1946–2014
v2x.polyarchy	Electoral democracy index (0–1)	V-Dem v15	1946–2014
instab	Political instability (large Polity swing or coup)	V-Dem v15	1946–2014
lgdp	Log GDP per capita (2011 PPP int'l \$)	V-Dem v15 (e.gdppc)	1946–2014
lpop	Log population (thousands)	V-Dem v15 / NMC v6 backup	1946–2014
oil	Oil/gas income per capita > 0 (binary)	V-Dem v15	1946–2014
cinc	Composite Index of National Capability	COW NMC v6	1946–2014
milex	Military expenditure (thousands USD)	COW NMC v6	1946–2014
milper	Military personnel (thousands)	COW NMC v6	1946–2014
major_power	COW major power status (binary)	COW Major Powers 2024	1946–2014
is_P5	UN Security Council P5 member (binary)	Coded from COW ccodes	1946–2014
ideal_point	UNGA ideal point (liberal–conservative)	Bailey et al. (2017)	1946–2023
recent_int	Military interventions by this state in prior 5 years	Regan (2000)	1946–1999
lmtnest	Log(% mountainous terrain + 1)	Fearon and Laitin (2003)	time-invariant
ncontig	Non-contiguous state (islands, exclaves)	Fearon and Laitin (2003)	time-invariant
colbrit	Former British colony	Fearon and Laitin (2003)	time-invariant
colfra	Former French colony	Fearon and Laitin (2003)	time-invariant
ethfrac	Ethnic fractionalization index (0–1)	Fearon and Laitin (2003)	time-invariant
eth_excl_frac	Population share of excluded ethnic groups	Cederman et al. (2010)	1946–2021
nwstate	New state (independent ≤ 2 years)	COW States 2011	1946–2014
prior_war	Ongoing civil war in previous year	COW ISW v5.1	1946–2014

Table 11: Country-year variables.

Variable	Description	Source	Coverage
ud_defense	Defense alliance (binary)	COW Alliances v4.1 (Gibler, 2009)	1946–2012
ud_entente	Entente alliance (binary)	COW Alliances v4.1	1946–2012
ud_A.biImports	Host’s imports from intervener (millions USD)	COW Trade v4.0	1946–2014
ud_log_capdist	Log capital distance (km)	COW capdist	static
ud_conttype	Contiguity type (1=land, ..., 6=none)	COW Contiguity v3.2 (Stinnett et al., 2002)	1946–2016
ud_peace	Peace quality score (0–1)	Diehl et al. (2021)	1946–2015
ud_icow_nclaims	Number of active territorial claims	ICOW v10.1 (Huth and Allee, 2003)	1946–2001
igo_shared	Shared full IGO memberships (count)	COW IGO v3 (Pevehouse et al., 2020)	1946–2014
ideal_point_distance	Absolute UNGA ideal-point difference	Derived from <code>ideal_point</code>	1946–2023
doe_pr_win_A	$P(A \text{ wins bilateral dispute})$	Carroll and Kenkel (2019)	1946–2012
A_wasColOf_B	A was a colony of B (binary)	COW coldata	static
B_wasColOf_A	B was a colony of A (binary)	COW coldata	static
ud_sharedColonizer	Shared former colonizer (binary)	COW coldata	static
ud_sameFirstEth	Same dominant ethnic group (binary)	Ellingsen (2000)	1945–2002
log_capratio	$\text{Log}(\text{cinc}_B / \text{cinc}_A)$ capability ratio	Derived from NMC v6	1946–2014
spat_gov	Polity-weighted fraction of other interveners: gov-biased	Derived (see App. E)	1946–1999
spat_opp	Polity-weighted fraction: opp-biased	Derived (see App. E)	1946–1999
spat_US_G	USA is gov-biased in this conflict (binary)	Derived	1946–1999
spat_USSR_G	USSR/Russia is gov-biased (binary)	Derived	1946–1999
spat_US_O	USA is opp-biased (binary)	Derived	1946–1999
spat_USSR_O	USSR/Russia is opp-biased (binary)	Derived	1946–1999
spat_US_USRG	B=USA and USSR gov-biased (binary)	Derived	1946–1999
spat_US_USRO	B=USA and USSR opp-biased (binary)	Derived	1946–1999
spat_USR_USG	B=USSR and USA gov-biased (binary)	Derived	1946–1999
spat_USR_USO	B=USSR and USA opp-biased (binary)	Derived	1946–1999

Table 12: Directed-dyad-year variables (B = potential intervener, A = host).

$$\text{spat_gov}_B = \sum_{\substack{j \neq A \\ j \neq B}} W_{Bj} \cdot \hat{\sigma}_j^*(1),$$

where $\hat{\sigma}_j^*(1)$ is the Nash equilibrium probability that state j intervenes on behalf of the government, as produced by the fixed-point iteration described in Section 2. An analogous expression gives spat_opp using $\hat{\sigma}_j^*(2)$.

A second pair of lags replaces the polity-similarity weighting with a *contiguity* weighting, capturing the strategic coupling of states that border the same conflict. Let $c_i = 1$ if potential intervener i shares a land border with the host A and $c_i = 0$ otherwise. The contiguity-chain weight matrix is $W^{\text{chain}} = c c^\top$ with zeroed diagonal and row normalization, so that for a host-bordering intervener the lag is the mean predicted intervention propensity of the *other* host-bordering states, and for a non-bordering intervener it is zero:

$$\text{spat_chain_gov}_B = \sum_{j \neq B} W_{Bj}^{\text{chain}} \hat{\sigma}_j^*(1),$$

with spat_chain_opp defined analogously from $\hat{\sigma}_j^*(2)$. Because states that border the same host are directly exposed to the same conflict, their intervention choices are strategically coupled through this classic conflict-diffusion channel. These two lags and the ten polity-similarity and superpower lags make up the 12 spatial features.

In the first training iteration (initialisation), $\hat{\sigma}_j^*(k)$ is replaced by the observed indicator $\mathbb{1}[y_j = k]$, where $y_j \in \{0, 1, 2\}$ is the Regan-coded intervention outcome. Subsequent iterations substitute the model’s own out-of-fold predictions until convergence, at which point the lags are self-consistent with the predictions they condition on.

F Stage 1 Classifier Diagnostics

This appendix provides the extended diagnostic output for the Stage 1 super-learner that is summarised in the main text. All statistics are out-of-fold unless stated otherwise; all figures and tables average across the 25 complete imputation datasets (5 country-year draws $\times$ 5 dyad draws) with standard deviations in parentheses or as error bars.

Learner	PRL (%)			NNLS weight (%)		
	X	W	XW	X	W	XW
Random forest	3.1	-130.1	8.3	32.0	0.6	20.4
MLP (100, 50)	25.4	1.1	22.9	11.5	0.2	6.4
HGB (lr=0.05)	18.7	11.2	17.7	8.8	0.7	5.2
Multinomial	28.3	13.6	28.9	2.4	1.0	3.6
HGB (lr=0.10)	-120.6	-57.3	-166.7	1.0	1.0	0.6
MLP (25)	-15.6	-186.3	-15.7	1.4	0.0	0.8
Ridge	29.6	13.6	29.8	0.1	0.7	0.7
Lasso	31.5	13.6	31.0	0.0	0.8	0.0
Elastic net	30.8	13.6	30.5	0.0	0.0	0.0
Super-learner	39.6			100		

Table 13: Out-of-fold PRL and NNLS weight for the 27 candidate models (9 learners $\times$ 3 feature sets) and the super-learner, averaged across the 25 draws. X = dyad features; W = spatial lags; XW = both. Sorted by total weight.

F.1 Component Learner Performance and Weights

Each of the nine component classifiers is trained on three feature sets: X (dyad characteristics only, no spatial lags), W (spatial lags only), and XW (both), yielding 27 candidate models. The NNLS stacking layer assigns weights to all 27 jointly. Table 13 reports out-of-fold PRL and NNLS weights for every learner broken out by feature set. Total weight by feature set: X = 57.3% (SD 12.8%), XW = 37.7% (SD 12.9%), W = 5.0% (SD 3.3%).

F.2 Calibration

Because Stage 1 probabilities are summed across approximately 135 dyads to form the country-year shadow measure, a calibration error of δ per dyad compounds to roughly 135δ in the aggregate—so even small miscalibration matters. Figure 2 presents reliability diagrams for each outcome class, pooled across all 25 imputation draws. In each panel, observations are sorted by predicted probability and divided into 10 quantile bins; the observed frequency within each bin is plotted against the mean prediction. The super-learner is well calibrated across all three classes: points track the diagonal closely,

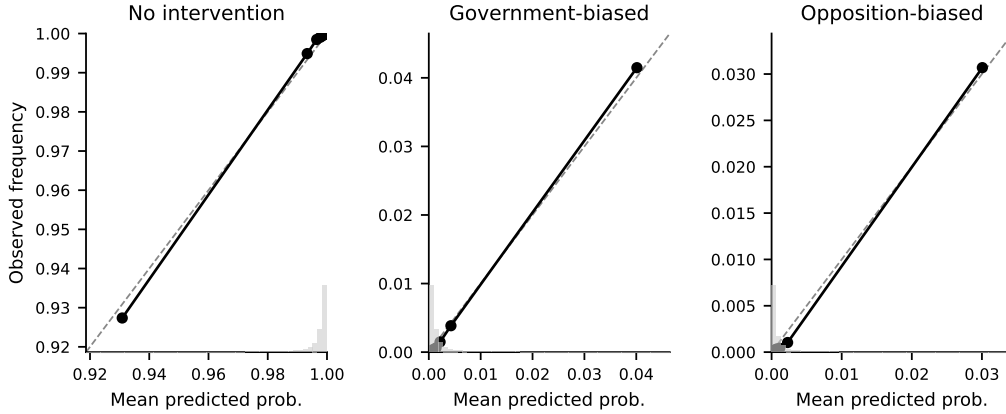


Figure 2: Reliability diagrams for each outcome class (no intervention, government-biased, opposition-biased). Each panel plots observed frequency against mean predicted probability in 10 quantile bins, pooled across all 25 imputation draws; the dashed diagonal represents perfect calibration. Axes are zoomed to the range where data reside. Shaded histograms show the distribution of predicted probabilities.

with no systematic over- or under-prediction.

F.3 Fixed-Point Convergence

This appendix documents a Stage 1 *training* diagnostic—whether the spatial lags fed to the classifier are self-consistent with the probabilities it predicts. It is distinct from the universal Nash fixed point that constructs the country-year shadow (Section 2), which operates over all directed dyads rather than the training sample.

F.3.1 The Consistency Requirement

The spatial lags `spat.gov` and `spat.opp` enter the Stage 1 classifier as predictors: they summarize how other potential interveners in the same conflict are behaving, weighted by ideological similarity. But the classifier itself produces predicted intervention probabilities, and it is those probabilities—rather than the realized 0/1/2 outcomes—that should, in principle, enter the spatial weights. The reason is that observed outcomes are single draws from

mixed-strategy equilibrium probabilities; a government-biased intervention (outcome 1) carries no information about whether the intervening state was playing a pure or mixed strategy.

Formally, the requirement is *self-consistency*: the spatial lags fed into $\hat{M}$ should be the weighted sums of the probabilities that $\hat{M}$ itself assigns to each dyad. Let $\hat{\mathbf{p}} = \hat{M}(\mathbf{X}(\mathbf{s}))$ be the vector of predicted probabilities, where $\mathbf{X}(\mathbf{s})$ denotes the feature matrix with spatial lags computed from $\mathbf{s}$. Self-consistency requires $\mathbf{s} = W\hat{\mathbf{p}}$, a fixed-point condition.

F.3.2 Diagnosis of Non-Convergence

The main training loop (Section 2) runs five rounds of retrain-then-update, initializing $\mathbf{s}^{(0)}$ from the realized binary outcomes. Whether five rounds reach self-consistency is measured by the residual at the five-iteration approximation $\mathbf{s}^{(5)}$: the mean absolute gap between $\mathbf{s}^{(5)}$ and the weighted predictions $W\hat{\mathbf{p}}(\mathbf{s}^{(5)})$ it should equal, which is exactly the correction the first burnout pass applies:

$$\Delta^{(5)} = \frac{1}{n} \sum_i |[W\hat{\mathbf{p}}(\mathbf{s}^{(5)})]_i - s_i^{(5)}|.$$

Averaged across all onset rows, both lag channels, and the 25 draws, $\Delta^{(5)} \approx 0.004$ —roughly 40 times the 10^{-4} self-consistency threshold. Five training rounds do not, on their own, reach self-consistency.

F.3.3 Frozen-Model Burnout

Retraining the full super-learner is computationally expensive (~ 3 hours per draw). An alternative is *frozen-model burnout*: after training is complete, hold the fitted $\hat{M}$ fixed and iterate spatial lags using its predictions until convergence. Each burnout pass costs only a prediction sweep (milliseconds versus hours).

The burnout algorithm is:

1. Initialize from the five-iteration approximation $\mathbf{s}^{(5)}$.
2. Apply $\hat{M}$ to features $\mathbf{X}(\mathbf{s}^{(k)})$ to obtain $\hat{\mathbf{p}}^{(k)}$.
3. Update $\mathbf{s}^{(k+1)} = W\hat{\mathbf{p}}^{(k)}$.
4. If $\max_i |s_i^{(k+1)} - s_i^{(k)}| < \tau$, stop; otherwise go to step 2.

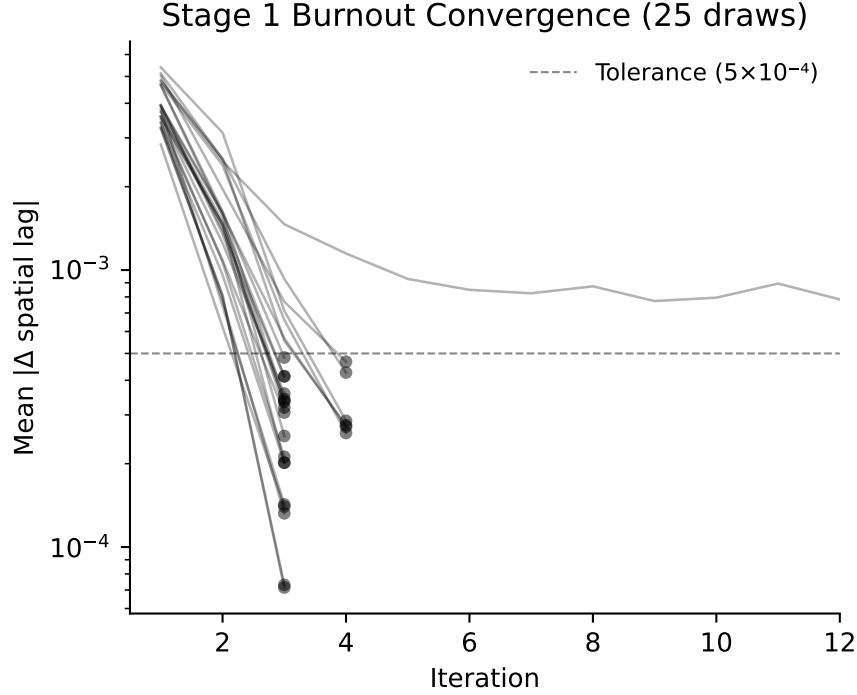


Figure 3: Burnout convergence across all 25 imputation draws. Each line traces the per-iteration mean $|\Delta_{\text{spatial lag}}|$ from the five-iteration approximation to convergence; dots mark endpoints. Dashed line: convergence tolerance (5×10^{-4}).

The tolerance is set at $\tau = 5 \times 10^{-4}$. This is above the theoretical 10^{-4} threshold because random-forest predictions have intrinsic stochasticity (different prediction calls can return slightly different probabilities for the same input due to tree-averaging over bootstrap samples). Empirical experiments confirm that Δ plateaus around 5×10^{-4} and cannot be reduced further without retraining.

Across all 25 imputation draws, burnout converges in 2–3 iterations (mean 2.2), with final Δ ranging from 0.7×10^{-4} to 4.8×10^{-4} —confirming that the plateau reflects irreducible prediction noise rather than failure to converge. Figure 3 shows the convergence trajectories: all draws exhibit rapid geometric decay (roughly halving each iteration) before plateauing at the tolerance threshold.

F.3.4 Interpretation: Approximate Equilibrium Selection

The burnout iteration does not find an equilibrium of the underlying intervention game; it finds a fixed point of the *fitted classifier* $\hat{M}$. The procedure is a nonparametric instance of the nested pseudo-likelihood (NPL) algorithm of Aguirregabiria and Mira (2007): where A&M iterate between estimating structural parameters and updating choice probabilities, the present application holds the fitted $\hat{M}$ fixed and iterates only the spatial lags (choice probabilities), corresponding to A&M’s two-step PML with a nonparametric first stage. These coincide with the true equilibrium only if $\hat{M}$ exactly recovers the true equilibrium mapping, which no finite-sample estimator can guarantee.

The weaker, defensible claim is as follows. Realized outcomes are draws from *some* equilibrium of the underlying game—they satisfy Nash consistency by assumption, since they were generated by the players. But the equilibrium correspondence for a game of this scale (up to 160 potential interveners, each choosing a mixed strategy) generically admits multiple equilibria, and different conflict cases may be near different connected components. The realized binary outcomes provide only coarse information about which equilibrium each case is near: a 0/1/2 outcome is consistent with any mixed strategy that places positive probability on that action.

The burnout iteration uses the model to extract additional information. Given the realized action as a starting condition, iterating the model refines the estimated spatial lags toward values that are approximately consistent with the smooth probability distribution implied by $\hat{M}$. In this sense, the iteration performs *approximate equilibrium selection*: it identifies, for each conflict case, which region of the equilibrium correspondence the realized outcome is most consistent with, according to the model’s learned mapping.

Three caveats are important. First, the model is trained on spatial lags derived from realized outcomes (the five-iteration warm start), not from the burnout-converged values. Retraining on the converged values would raise a circularity concern: the model’s weights would depend on predictions the model itself generated. By freezing the model before burnout, the learned coefficients remain anchored to the training-time feature distribution, and the converged lags represent a small perturbation ($\Delta \approx 0.003$ relative to the warm start) rather than a wholesale shift to a foreign equilibrium. Second, the equilibrium selected for each training case by the burnout need not be the same equilibrium that governs prediction-time cases. The identify-

ing assumption is that the equilibrium selection mechanism is stable across the sample—the same forces that push cases toward particular equilibria in training also govern unobserved cases. Third, the burnout-converged lags are used only to compute the shadow measure (the country-year expectations E_{gov} and E_{opp} in Stage 2) and not to re-estimate Stage 1. The shadow measure therefore reflects the approximately-equilibrium spatial environment, while the classifier’s weights remain from training on the realized-outcome initialization.

G Properties of the Shadow Measure

This appendix documents the descriptive behavior of E^G and E^O , summarized in Section 2.6 of the main text.

Before turning to the onset analysis, I examine the descriptive properties of E^G and E^O to establish that they behave as the theory predicts.

The dyad-level predictions reveal a structural asymmetry between the two sides of intervention. The highest predicted probabilities of government-biased intervention involve superpowers projecting influence into client states—the United States supporting the Philippines in 1972 ($\hat{p}_{\text{gov}} = 0.65$), Laos in 1962, and Cambodia in 1971. For opposition-biased intervention, the picture shifts: the maximum is Israel opposing the Syrian government in 1982 ($\hat{p}_{\text{opp}} = 0.57$), followed by Bulgaria targeting Greece in 1946 and Kuwait targeting Iraq in 1991, and nine of the ten highest-ranked opposition predictions involve non-P5 interveners. A measurement framework restricted to P5 states would miss the majority of the opposition-biased shadow.

Table 14 makes this concrete for three canonical onsets, reporting the five states to which the classifier assigns the highest government- and opposition-biased intervention probabilities alongside an indicator of whether the state actually intervened in that direction.

Across the three cases, 11 of 13 actual interventions appear in the top five on the correct side—and in each case, restricting the measure to P5 states would discard the single most important opposition-biased intervener: Zaire in Angola, Somalia in Ethiopia, Pakistan in Afghanistan.¹¹

¹¹The United States appears in the top five government-biased predictions for all three cases, yet actually intervened on the opposition side in Angola and Afghanistan. The model assigns substantial probability mass to *both* sides but defaults toward government-biased intervention because that is the modal US posture across the full sample. This directional

Gov-biased	$\hat{p}$	Opp-biased	$\hat{p}$
Angola (1976) — 5 actual interventions			
United States*	.26	Zaire	.49
Cuba	.15 ✓	South Africa	.21 ✓
USSR*	.15 ✓	United States*	.04 ✓
Congo	.07	Congo	.04
Zambia	.07	Portugal	.04
Ethiopia (1978) — 4 actual interventions			
USSR*	.17 ✓	Somalia	.39 ✓
Cuba	.16 ✓	United States*	.07
United States*	.14	North Yemen	.06
Kenya	.05	Sudan	.04
China*	.04	Cuba	.02
Afghanistan (1978) — 4 actual interventions			
USSR*	.15 ✓	Pakistan	.20 ✓
China*	.09	China*	.08
United Kingdom*	.09	Iran	.07 ✓
Iran	.09	United States*	.03 ✓
United States*	.05	United Kingdom*	.02

Table 14: Top five predicted interveners for three canonical onsets; out-of-fold probabilities averaged across the 25 imputation draws. *P5 member; ✓ the state actually intervened in that direction (Regan coding).

Figure 4 shows the cross-sectional distribution of E^G and E^O across all country-years. Both measures are heavily right-skewed: the median country-year faces modest intervention expectations, but a long right tail captures the handful of states embedded in dense alliance networks and superpower competition.

Figure 5 traces the temporal dynamics for three of the most heavily internationalized civil wars of the Cold War. Beyond magnitude, the shadow recovers *direction*—which outside powers line up on which side. Angola is

ambiguity for superpowers is one reason the country-year aggregation is preferable to raw dyad-level predictions: summing across all potential interveners preserves the correct signal that these countries faced unusually high total intervention pressure, even where the classifier hedges on which side a specific superpower would take.

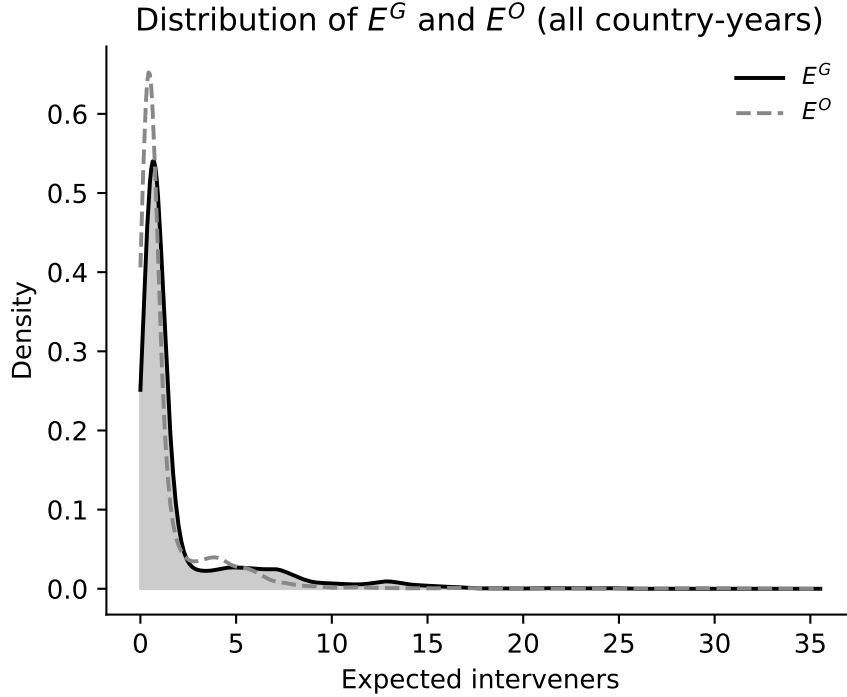


Figure 4: Kernel density estimates of E^G and E^O across all country-years (1946–2014), averaged across 25 imputation draws.

the clearest case: at the 1976 onset the government-side shadow ($E^G = 1.52$) is anchored by Cuba and the Soviet Union, the MPLA’s patrons, while the opposition-side shadow ($E^O = 1.16$) loads on Zaire, South Africa, and the United States—the external backers of the FNLA and UNITA. Ethiopia shows the same structure through the 1978 realignment: Cuba and the Soviet Union again anchor the government side for the Derg ($E^G = 1.04$), while Somalia drives the opposition side in the Ogaden war. Afghanistan illustrates a regime switch: the 1978 Saur Revolution lifts the government-side shadow, but by 1979 the opposition side overtakes it ($E^O = 0.75$ vs. $E^G = 0.94$), capturing the surge of mujahideen support from Pakistan, Iran, and the United States against a Soviet-backed government. In every case the shadow concentrates on the historically correct interveners rather than diffusing across the system—evidence that it tracks the strategic environment, not merely the onsets on which Stage 1 is trained.

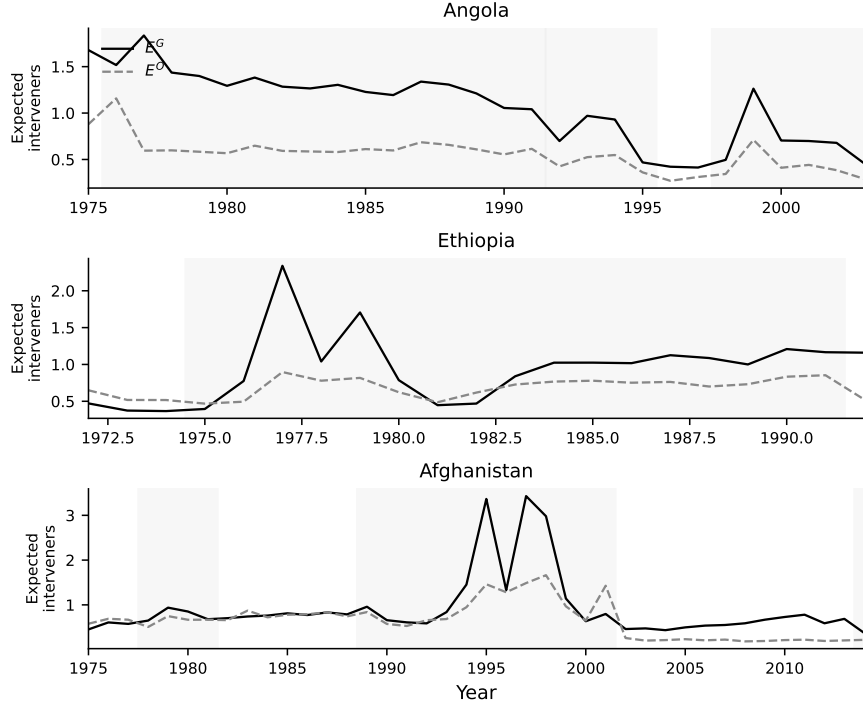


Figure 5: Shadow measure time series (E^G and E^O) for three countries with well-documented intervention environments. Probabilities averaged across 25 imputation draws.

E^G correlates positively with both existing measures of anticipated intervention: the Lake security hierarchy used by [Cunningham \(2016\)](#) ($r = 0.29$, $n = 162$) and the structural P5 intervention probabilities estimated by [Gibilisco and Montero \(2022\)](#) ($r = 0.47$, $n = 146$). The stronger correlation with G&M is expected—both capture intervention propensity directly—while the moderate hierarchy correlation reflects the fact that US dominance is only one component of the broader intervention environment.

H Full Second-Stage Estimates

Table 15 reports the complete *Entrants* onset model whose two shadow coefficients are summarized, in the identified net-tilt parameterization, in Table 6. Coefficients and intervals are computed by the $T \times P$ procedure, so the inter-

	Coef.	95% $T \times P$ interval
γ^G (gov.-biased shadow)	-1.61	[-6.14, +0.43]
γ^O (opp.-biased shadow)	+0.19	[-2.53, +1.97]
Constant	-12.01	[-38.25, +14.28]
Democracy	-0.03	[-0.06, -0.00]
Log GDP per capita	-0.44	[-0.79, -0.11]
Log population	+0.12	[-0.08, +0.31]
Mountainous terrain	+0.13	[-0.09, +0.31]
Noncontiguous	+0.51	[-0.18, +1.12]
Oil exporter	+0.25	[-0.27, +0.79]
New state	+1.20	[+0.22, +2.00]
Political instability	+0.71	[+0.29, +1.22]
Prior war	+0.75	[+0.15, +1.39]
Ethnic frac.	+0.50	[-0.28, +1.37]
Religious frac.	+0.93	[-0.21, +2.07]
Year trend	+0.00	[-0.01, +0.02]

Table 15: The full *Entrants* onset model. Coefficients are $T \times P$ means; intervals are the 2.5th–97.5th percentiles. Net tilt = $(\gamma^O - \gamma^G)/2$, common intensity = $(\gamma^O + \gamma^G)/2$.

vals propagate the shadow’s measurement uncertainty and are correspondingly wide for the collinear shadow channels; the controls are conventionally estimated.

I Predictive-Significance Tests

Table 3 scores each covariate by what its removal costs a held-out forecast. This section reports the formal counterpart: the prediction-difference test of [Mentch and Hooker \(2016\)](#), the procedure [McAlexander and Mentch \(2020\)](#) bring to onset data. Trees are built on subsamples ($k = 75$ of $n = 8,835$; minimum split 3), so ensemble predictions are asymptotically normal U-statistics with estimable covariance; the test compares an ensemble built on the observed data against one built with the focal columns permuted, evaluating both at N held-out test points and referring $\hat{D}'\hat{\Sigma}_D^{-1}\hat{D}$ to χ_N^2 . Following the original authors’ discipline, the tuning was validated on calibration anchors

before any shadow test was run: across 20 pure-noise focal columns the p -values are near-uniform (mean 0.58; 1 of 20 at or below 0.05, the nominal rate), and a synthetic strong signal is rejected decisively ($\chi^2_{20} = 347$).

Verdicts at any single set of test points are variable in a rare-event outcome—the original authors note the test’s sensitivity to test-point location—so the joint test of the two shadow channels, the appropriate test for a collinear pair, was run across ten independent test-point draws (five each at $N = 20$ and $N = 40$) and combined by twice the median p -value (Meinshausen et al., 2009). It rejects at every configuration: 9 of 10 draws at $p \leq 0.005$, all ten at $p \leq 0.03$, combined $p \approx 0.001$ at both test-set sizes. The marginal tests are configuration-sensitive in exactly the way collinearity predicts—each channel’s verdict depends on which test points happen to separate it from its twin—the same pathology McAlexander and Mentch (2020) document for the fractionalization measures in the Fearon–Laitin model, and the empirical motivation for working in the tilt and intensity combinations of Section 3.3. The structural controls do not reject (income $p = 0.47$, population $p = 0.93$).

A conventional permutation counterpart agrees: refit the ensemble under each of 99 draws from the focal columns’ permutation null, score each refit out of fold, and compare the observed average precision to that null—the brute-force analogue of the computationally efficient test of Coleman et al. (2022). This version permits the generated-regressor honesty the asymptotic machinery does not contemplate: run separately on each of the 25 measurement draws, the joint test rejects at $p \leq 0.05$ in every draw (median 0.01, worst 0.03), and the marginals clear conventional thresholds in most draws (E^G in 22 of 25, E^O in 19) while remaining draw-sensitive. The measurement spread does not rescue the null.